



Research on insulator fault detection and recognition based on deep learning in complex weather conditions

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ABSTRACT

As an important part of the transmission chain, insulators, together with other key components, determine the reliability of the system. Furthermore, in the context of complex weather conditions the recognition of images may prove challenging. Consequently, it is of paramount importance to investigate insulator fault detection methodologies in the context of complex weather to enhance the precision of insulator fault detection. This study presents two new methods: the FC-NET method for image dehazing based on feature fusion, and the TFRAIN method for image deraining based on mixed-scale Transformer. Furthermore, the YOLOv7 network is proposed as the foundation for a fault detection network, designated as YOLOv7-EP. This network incorporates a multi-scale attention mechanism and is constructed as an integrated model for insulator detection in complex weather conditions through the combination of image processing techniques. The results of the experiments demonstrate that the FC-NET method exhibits a PSNR of 10.3% higher and an SSIM of 2.4% higher than the current optimal method, while the TFRAIN method also demonstrates a PSNR of 10.3% higher and an SSIM of 2.4% higher than the current optimal method. The validation of the new fault detection network, YOLOv7-EP, through comparative and ablation experiments revealed an overall parameter reduction of 11.6%, a computational reduction of 18.3%, and an accuracy improvement of 0.7%.

1. Introduction

The reliability of power transmission systems relies on the synergy of several components, including conductors, insulators and support structures. Among them, insulators, as important electrical insulation components in power equipment, are widely used in power transmission chain [1]. Due to their prolonged exposure to a multitude of intricate environments, these components are susceptible to a plethora of deleterious factors, including elevated temperatures, avian predation, wind erosion, and ice-covered pressure, which can culminate in a spectrum of failure modes [2]. Conducting a comprehensive investigation into the insulator fault detection mechanism is imperative, as is implementing efficacious preventive measures to guarantee the dependability of the power system. The limitations of traditional manual inspection include the constraints of experience, responsibility, and labour intensity, as well as the potential issues of leakage and safety hazards. The popularity

of drone inspection technology is due to its portability, low cost and high efficiency. However, the limitations of simple drone photography and manual analysis mean that this approach is no longer sufficient. The Edge Cloud Collaborative Power Inspection System employs a combination of cloud computing and edge intelligence to enhance inspection efficiency through the utilisation of AI and cloud resources. However, inclement weather conditions, such as fog or precipitation, can impair image quality and diminish the efficacy of the inspection process. It is therefore necessary for the intelligent inspection system to incorporate dehazing and deraining technologies to improve image quality, enhance insulator fault detection efficiency. Notwithstanding the constraints imposed by flight safety regulations and visibility on UAV inspections conducted in extreme weather conditions, there are numerous instances of light to moderate haze or light rain in most maintenance scenarios. It is evident, from the experiential knowledge of power grid equipment operators, that inclement weather does not invariably impede the

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execution of inspection procedures. Conversely, such conditions present a pivotal opportunity to expand the scope of inspection operations. Consequently, conventional image enhancement methodologies encounter difficulties in achieving an optimal balance between real-time performance and efficacy. However, the joint framework proposed in this study has the capacity to rapidly and efficiently restore image details while ensuring the UAV flies safely, thereby maintaining the identifiability of small fault targets. This development has the potential to significantly expand the scope for inspections, thereby reducing the prevalence of blind spots in relation to weather-related conditions. It is anticipated that this enhancement will improve the continuity and reliability of online maintenance for power grids.

2. Related Works

2.1. Image quality optimization technology

2.1.1. Image dehazing methods

In recent years, image dehazing has emerged as a pivotal challenge in the domain of image processing, garnering significant attention from the research community. The research can be broadly classified into two categories: traditional image dehazing algorithms and those based on deep learning. Each approach has its own merits. The primary objective of traditional image dehazing algorithms is to enhance the contrast and clarity of foggy images. Since the initial research conducted by L. Bissionnette et al. [3] in 1992, numerous advancements have been made in the field of image dehazing, including the development of image enhancement and image prior-based dehazing methods [4–6]. However, traditional algorithms often face the problems of difficult parameter selection and unstable effect when dealing with complex scenes and dynamic environments.

With the advent of deep learning, image dehazing algorithms based on deep learning have increasingly become the dominant approach. These methods identify the mapping relationship between the two by training a dataset comprising both fogged and unfogged images, and extracting deep features using convolutional neural networks (CNNs). Cai et al. [7] introduced DehazeNet for the first time and achieved enhanced results by learning transmission transmittance maps. Subsequent developments in the form of multi-scale network models and a densely connected pyramid dehaze network (DCPDN) have further enhanced the dehazing effect [8,9]. End-to-end networks that do not employ atmospheric scattering models have also demonstrated notable advancements in single-image dehazing [10]. For instance, networks based on encoder-decoder architectures coupled with contrastive learning have attained remarkable dehazing outcomes through the utilisation of guided transmission maps [11,12]. Transformers have been instrumental in advancing image dehazing methods. The DehazeFormer model has been proposed as a solution to the image denoising problem, a problem which is addressed by combining Swin Transformer and U-Net techniques [13–15].

Despite the efficacy of algorithms such as DehazeNet and DCPDN in restoring blurred images, there is a paucity of specialised optimisation for features such as the microscopic texture and reflectance of insulator surfaces. Moreover, their computational overhead remains excessively high in real-time environments, necessitating further improvement.

2.1.2. Image deraining method

The majority of image deraining methods can be classified into two principal categories: those that utilise traditional model optimisation and those that employ data-driven deep learning. Traditional methods, including sparse coding, low-rank matrix recovery, and Gaussian mixture models, achieve the separation of rain and background layers by examining the physical properties and a priori knowledge of rain streaks [16,17]. In recent years, there has been a notable advancement in deep learning-based deraining methods. The methodologies proposed in the

2017 research article on the Rain1400, Rain100H and Rain100L datasets have enhanced the capabilities of deep residual networks, thereby facilitating the advancement of research in this field [18]. Nevertheless, fully supervised or semi-supervised learning paradigms remain inadequate in real-world scenarios [19]. Transformer has been applied to image restoration following its success in natural language processing and advanced vision tasks. Vision Transformer (ViT) segments an image into image chunks and converts them. The image chunks are then fed into the Transformer for processing [20]. IPT was the first to apply ViT to image deraining research [21], while Uformer and Restormer have demonstrated the ability to efficiently capture long-range and complex rain features through improved network architectures [22,23]. Although these deep learning methods significantly outperform traditional algorithms in terms of dehazing effectiveness, they still require large amounts of labelled data, which can be a major challenge in practical applications.

It has been demonstrated that traditional sparse coding and low-rank recovery methods are susceptible to detail loss in scenarios involving heavy rain. Despite the capacity of deep networks (The Rain1400 series, Uformer and Restormer) to capture long-range rain threads, they are deficient in the scene adaptation and lightweight design necessary for UAV aerial photography of insulators.

2.1.3. Other image enhancement techniques

In recent years, there have been many breakthroughs in the field of severe weather image restoration: Jin et al. [24] transformed the image restoration problem into higher-order coefficient estimation based on Taylor expansion and multiscale attention mechanism, and realised fine correction of complex atmospheric scattering, providing a theoretical reference for mixed-scale feature fusion; The spatio-temporal interaction module proposed by Zhang et al. [25] shows unique advantages in dynamic rain pattern modelling. Zhang et al. [26] propose a Deep Dense Multi-Scale Network (DDMSNet) for snow removal by exploiting semantic and depth priors to learn a semantically and geometrically aware representation via a self-attention mechanism to produce the final clean images. The process of snow and hail image removal is very similar to rain and fog removal in terms of model structure and damage design, relying on multiscale feature extraction and attention mechanisms to distinguish weather disturbances from scene details.

The efficacy of classical enhancement methods is constrained by their inability to adequately improve overall contrast and their failure to enhance small targets, such as microcracks and stains, to a satisfactory degree. The development of multi-scale and attention mechanism frameworks has introduced novel approaches to address these limitations; however, these frameworks have yet to consider the requirements for real-time deployment.

2.2. Visual inspection method

2.2.1. General object detection algorithm

In the field of power systems, the detection of insulator faults is typically conducted through the use of unmanned aerial vehicle (UAV) aerial photography. Deep learning-based methods demonstrate significant advantages in the context of fault detection tasks. The target detection algorithms can be classified into two main categories: two-stage and one-stage. Two-stage algorithms, exemplified by R-CNN and Fast R-CNN, facilitate defect detection through the generation of candidate regions, which are subsequently classified [27–31]. One-stage algorithms, such as the YOLO series and the SSD series, directly regress the input image and output the results. In comparison to other target detection algorithms, YOLOv7, a popular variant of the YOLO series, demonstrates superior accuracy, speed, scalability and generality [32–36]. Although these methods perform well in static backgrounds, their accuracy and speed are still limited in dynamic environments, especially in complex weather conditions. However, the existing YOLO series of methods still face the problem of image quality degradation

when dealing with images under severe weather conditions, which affects the accuracy of fault detection.

2.2.2. Specialized method for detecting insulator defects

In terms of insulator detection, with the development of deep learning and image processing techniques, insulator fault detection techniques have made significant progress. Fan et al. [37] proposed a fast non-contact detection method using hyperspectral imaging (HSI), which achieved a classification accuracy of 96.81 % by extracting hyperspectral spectral lines and features of the samples combined with a random forest (RF) model, and it provides an efficient engineering tool for assessing the aging state of insulators. Meanwhile, Belhouchet [38] optimized the design of corona rings on insulators using a design of experiments approach and the Bat algorithm, which significantly reduced the electric field strength and increased the reliability of the system. These studies have achieved technological breakthroughs and demonstrated remarkable results in practical applications. In addition, Zhang et al. [39] proposed a method to assess the pollution level of insulators based on discharge noise, and used a probabilistic neural network (PNN) to analyze the effects of different pollution levels, ambient temperatures and relative humidity on the characteristic parameters of insulator discharge noise, which demonstrated the potential of the method in real-time non-contact pollution assessment. Finally, Aleena Swetapadma and Anamika Yadav [40] proposed an EfficientNet method based on transfer learning that achieved 99.9 % classification accuracy by extracting features from insulator images, significantly improving the efficiency and accuracy of insulator defect detection. The innovation of this approach is the first application of EfficientNet to insulator defect detection, demonstrating its potential in image processing. Despite the advances in existing insulator detection techniques, there are still limitations. Traditional methods rely on manual inspection and simple image analysis, which makes it difficult to maintain accuracy under complex weather conditions, especially in foggy and rainy days when image quality degradation affects fault detection. In addition, deep learning models usually require a large amount of labelled data, which is difficult to achieve in practical applications and underperforms in small target detection and variable weather conditions. Therefore, future research should focus on integrating advanced techniques to improve detection performance and reliability.

The detection of insulator faults presents a significant challenge in the context of intelligent inspection. While existing methods are effective, the accuracy and speed of detection models are of greater importance in real-world inspections. Alahyari et al. [41] proposed a deep learning based two-stage model that achieves over 90 % segmentation and classification accuracy in normal weather and non-uniform backgrounds. The method uses U-Net for segmentation and VGG for classification, demonstrating the efficiency of the optimized model architecture under sunny weather.

In recent years, Hao et al. [42] proposed PKAMNet network, which integrates a priori knowledge migration and attention mechanism for transmission line insulator parallel gap fault detection, significantly improving the detection accuracy of small targets. However, these do not consider image preprocessing and target detection under complex weather conditions, and further consideration needs to be given to the robustness and adaptability of the model in multiple harsh environments. Therefore, the effect of complex weather on image acquisition must be further investigated to improve the comprehensive performance of fault detection systems. The techniques employed for the removal of fog and rain from images become particularly important in situations where inclement weather is a factor. Notwithstanding the considerable advances made in image dehazing technology, traditional algorithms remain constrained by limitations and the challenge of parameter selection in complex scenes and variable weather conditions. Deep learning-based methods, while effective, necessitate a substantial amount of labelled data and powerful computational resources. Furthermore, methods for deraining from images encounter difficulties

when confronted with intricate and diverse precipitation patterns, as well as the challenge of recovering image details. To address these issues, research must enhance the resilience and efficiency of fog and deraining algorithms, particularly in the context of insulator fault detection. In conditions of optimal visibility, specialised detection models (HSI + RF, PNN, EfficientNet and PKAMNet etc.) have been shown to achieve superior performance. However, there is often a neglect of image pre-processing and target enhancement coordination, which complicates achieving an equilibrium between the robustness of detection and the ability to recognise small targets in complex weather conditions.

Subsequent iterations of this system, including YOLOv10 and YOLOv11, have been released. Nevertheless, the present study employs YOLOv7 as its foundational framework for three principal reasons: Firstly, since its release in 2022, YOLOv7 has been thoroughly validated and proven to be robust, with abundant community resources and pre-trained weights, making it easy to reproduce and further develop. Secondly, it natively supports lightweight and multi-scale attention mechanisms (such as EMA and PConv), which align well with the improvements proposed in this paper. In contrast, the interfaces of the newer versions are still unstable and lack sufficient engineering practice cases. Thirdly, while YOLOv10/11 offers significant improvements in terms of accuracy, the YOLOv7 network is more straightforward and has reduced computational and memory overhead, rendering it more suitable for real-time small object detection on embedded systems or drones. Following a thorough evaluation of the available options, it was determined that YOLOv7 would serve as the foundation for this study. Although the experimental validation in this work is carried out only on rain and fog scenes, the proposed FC-NET, TFRAIN and YOLOv7-EP frameworks are in principle applicable to snow and hail image recovery and show good generalisability. Compared with the above methods, This study pioneered the introduction of the physical process-driven feature fusion mechanism, which effectively enhances the synergy of multi-scale features in the two stages of reduction and detection through physically guided a priori constraints in the multi-scale “first de-raining/fogging → re-detecting” cascade paradigm; At the same time, based on the requirements of the detection task, the lightweight model tailoring and at the same time, the lightweight model tailoring and acceleration design based on the requirements of the detection task significantly reduces the amount of model computation, ensuring fast deployment on embedded or edge devices; on multiple test sets with different rain/fog intensities, the framework achieves stable detection recall and visual restoration, which is better than the existing SOTA methods in terms of both accuracy and speed.

In this context, three principal issues have been identified: performance, reliability and computational efficiency. The following section outlines the contributions made by the authors of this work:

- (1) The proposed FC-NET method significantly improves insulator image denoising by feature fusion
- (2) The TFRAIN method based on mixed-scale transformers greatly improves insulator image capacity reduction.
- (3) The newly proposed YOLOv7-EP algorithm achieves lighter and more efficient detection of small targets and can improve enhance the performance of insulator detection under complex weather conditions.

In general, our proposed feature fusion de-fogging method, mixed-scale Transformer deraining method and YOLOv7-EP algorithm are more effective than existing techniques in terms of both qualitative and quantitative metrics. This is particularly evident in the detection of small targets in blurred backgrounds and low resolutions, which is suitable for insulator detection under complex weather conditions without increasing computational burden.

3. Materials and methods

To effectively address the challenges posed by the multiple interferences of complex meteorological conditions (e.g., foggy and rainy days) on the image quality of power inspection, this study proposes an insulator fault detection system for practical application scenarios. The system's overall design is predicated on the task-oriented principle of "enhancing image discrimination, extracting robust features, and improving fault identifiability". The architecture is built on an end-to-end collaborative basis and consists of two phases, namely image enhancement and target detection. The aim is to improve the accurate identification of small insulator faults (e.g., cracks, nicks, and stains) under complex weather conditions.

The system is comprised of multiple sub-modules, each with distinct functions, which are targeted and optimised for various weather degradation factors. In the image preprocessing stage, the feature fusion network FC-NET is proposed for the purpose of dealing with haze interference. The network utilises a pre-haze sub-module constructed by a deep convolutional network to generate a guiding image. Subsequently, the multi-scale contextual features of the original image and the guiding image are integrated through the feature fusion module, with the objective of achieving clear recovery of low-contrast areas. The TFRAIN module based on the transformer combines the window of the Lewin Transformer with the TFRAIN module. In the context of raindrop occlusion, the transformer-based TFRAIN module has been introduced. This module integrates the window attention mechanism of the Lewin Transformer with the mixed-scale feed-forward network (MSFN). It does so in a manner that considers both global context modelling and local texture detail restoration. The TFRAIN module has been shown to enhance the reconstruction of image quality in scenes such as rainstorms, reflections, and water-damaged disturbances. The enhanced image, which is characterised by its clarity, is utilised as the input to the back-end detection module. In the context of fault detection, this paper proposes an enhancement to the YOLOv7 network structure, namely the YOLOv7-EP model. This model integrates the multi-scale attention mechanism (EMA) and Partial Convolution (PConv) optimization strategy, thereby demonstrating the capacity to focus on small target faults (e.g. edge cracks, minor breakage) and achieving a substantial

enhancement in the model's detection accuracy and stability.

In summary, the integrated rain and fog removal enhancement-detection system constructed in this paper is structured around the main features of image degradation under complex weather conditions, forming a closed-loop task from image quality enhancement to insulator fault identification, and possesses strong practicability and deployment potentials while improving detection accuracy.

3.1. Dataset

This paper employs two aspects of the dataset. Firstly, insulator images in foggy scenarios are challenging to capture with the same time, angle and position in real production. Secondly, the presence of fog makes it impossible to collect real foggy and non-foggy images in the same state. This experiment utilises an insulator detection dataset, which is available in the literature, comprising approximately 13,700 generated foggy image pairs. Some examples of the foggy image data are shown in Fig. 1 [43].

Conversely, given the limited publicly available insulator dataset and the challenge of acquiring images with and without rain in the same setting, the experiment initially sought to augment the data set. The original data, comprising 600 normal images and 248 defective images, was selected from the China Power Grid Insulator Data (CPLID) set. The data was augmented to comprise 2544 images through the application of horizontal and vertical mirror flipping, with the locations of normal and defective insulators subsequently labelled using Label Img. In accordance with the methodology outlined in the relevant literature [44], which serves as a reference for the real-scene rainy day insulator dataset, the rainy-day data in the real scene was generated using Photoshop. This synthesis method is widely utilised and has proven to be highly reliable. This entailed the introduction of rain layers, assorted hues, motion blur, and Gaussian blur, in accordance with the prescribed steps. Ultimately, the angle and density of the rain stripes were adjusted to align the data more closely with real-world conditions, resulting in the generation of 2544 synthetic rain maps with varying rain textures. Fig. 2 illustrates some of the synthetic rain data.



Fig. 1. Illustrates an example of a partially synthesised fog map.



Fig. 2. Illustrates an example of a partially synthesised rain map.

3.2. This study presents a novel feature fusion-based dehazing method for insulator images

The current state of research in image de-fogging can be broadly classified into two methods. The first is the image prior-based de-fogging method, which primarily addresses haze by estimating transmittance and atmospheric light. However, this approach has limited efficacy in complex scenes. The second is the end-to-end trainable de-fogging network, which generates clear images directly from foggy images. However, this method lacks training guidance and is challenging to optimise. In order to enhance the precision of intelligent inspection of power systems in complex settings, a comprehensive investigation of the characteristics of foggy images is essential to refine the algorithm. To address these issues, a feature fusion-based image dehazing method is proposed. The proposed approach is inspired by SGID-PFF [45], which employs an end-to-end training strategy. Firstly, to identify and extract useful information from foggy images, a pre-fogging module is constructed based on a CNN. This module identifies pertinent

characteristics within the input foggy images, subsequently generating a bootstrap image to inform the subsequent de-fogging process. Subsequently, two distinct feature extraction techniques are employed to extract features from the guidance image and the fogged image, respectively. Thereafter, the feature fusion module is introduced to fuse the features extracted from the foggy image and the guiding image using a step-by-step feature aggregation method, thereby facilitating the restoration of the clear image. Ultimately, the image is dehazed using the fused features via the image recovery module, as shown in Fig. 3.

3.2.1. Pre-Dehazing network module

According to the atmospheric scattering model, the clear image $J(x)$ can be expressed as Eq. (1).

$$J(x) = I(x) \frac{1}{t(x)} + A - \frac{A}{t(x)} \quad (1)$$

In cases where both $t(x)$ and A are uncertain parameter values, it is of paramount importance to accurately estimate these parameters to

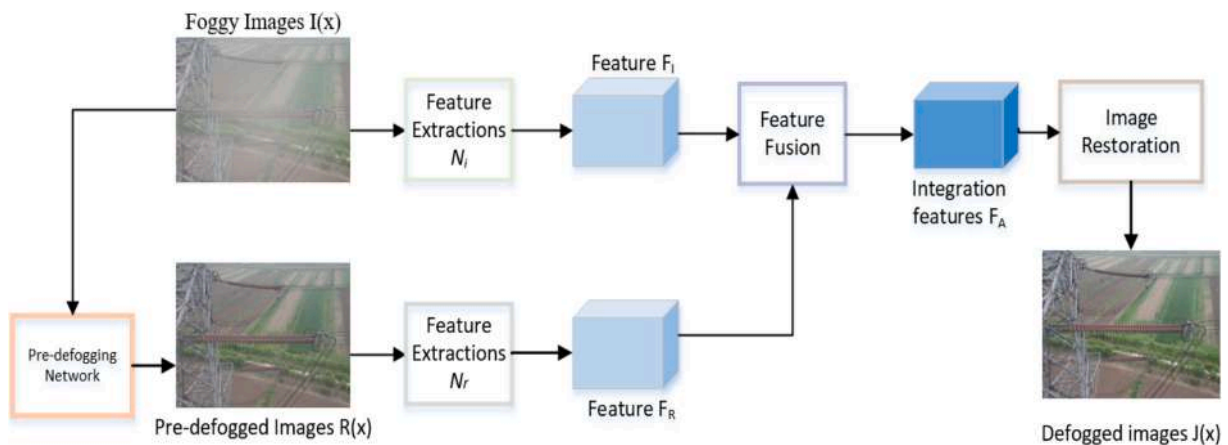


Fig. 3. Illustrates the detailed structure of the dehazing network.

obtain optimal dehazing results. In the majority of cases, atmospheric light A is assumed to be a global constant that affects solely the colour of the image recovery. Therefore, $A = 1$ can be assumed in order to perform image recovery without affecting the structure of the image itself; however, this will result in colour distortion of the image. In order to rectify this discrepancy, a correction factor, designated as $C(x)$, is introduced to address the inaccuracy introduced by the assumed value. The expression for the estimation process for a clear image is as Eq. (2).

$$J(x) = I(x) \frac{1}{C(x)} + 1 - \frac{1}{C(x)} \quad (2)$$

Then according to (2) and (2), the relationship between the correction coefficient $C(x)$, transmittance $t(x)$ and atmospheric light A can be derived as:

$$C(x) = \frac{t(x)(I(x) - 1)}{t(x)(A - 1) + I(x) - A} \quad (3)$$

Accordingly, the image dehazing task can be transformed into estimating the correction coefficient $C(x)$ from the fogged image in accordance with Eq. (2) without considering the values of $t(x)$ and A simultaneously. This approach simplifies the complexity of the dehazing problem.

From Eq. (3), it is evident that the correction coefficient $C(x)$ is associated with the transmittance $t(x)$. In the traditional a priori-based dehazing method, estimation of transmittance and atmospheric light is required to achieve an improvement in image dehazing. This is achieved by using certain universal properties in clear images to guide or optimise the estimation of transmittance maps through a priori information.

In the dehazing algorithm, a pre-dehazer network N_f is employed to generate the bootstrap image $R(x)$ from the fogged image. This is then used as a guide to estimate the correction coefficients $C(x)$. The N_f in the pre-fogger is a deep CNN [46]. The network is constructed based on an encoder-decoder architecture, and its network structure is shown in Fig. 4.

Each encoder and decoder comprise a 5x5 convolution or 3x3 anti-convolution block and three ResBlock blocks. Following the input of the fog image, the features are initially extracted through convolution and subsequently processed by ResBlock, with the dotted line representing the residual connection. The outputs of the encoder are added to the corresponding inputs of the decoder via jump connections. The convolutional and inverse convolutional layers employ ReLU activation functions, whereas the final convolutional layer outputs a $C(x)$ estimate utilising a Sigmoid activation function. The ResBlock block structure comprises two 5x5 convolutions and a ReLU activation function, which is designed to expand the receptive field, extract features and facilitate learning. In essence, the inputs are linked to the outputs via residuals,

thereby enhancing the modelling capabilities of the network. The detailed structure of the ResBlock block is shown in Fig. 5.

By applying the pre-fogging network described above, the expression for the generated bootstrap image $R(x)$ can be derived from Eq. (2), as presented in Eq. (4).

$$R(x) = I(x) \frac{1}{\mathcal{N}_f(I(x))} + 1 - \frac{1}{\mathcal{N}_f(I(x))} \quad (4)$$

The generated bootstrap image $R(x)$ contains structural information from the clear image and provides a reference for the subsequent dehazing process. However, due to the reliance on an approximation of atmospheric light in the pre-demisting process, the generated bootstrap image will typically exhibit a minor degree of residual fog and some discrepancies in colour.

In the design of pre-defogger network architecture for FC-NET, this paper adopts the encoder-decoder structure and configures a specific ResBlock module. The encoder part extracts multi-scale features by gradual downsampling, while the decoder part uses jump connections to recover spatial details step by step, effectively preserving the low-level texture information. Compared with the conventional U-Net or ResNet structure, FC-NET is specially optimized in the feature fusion mechanism, and the cross-scale feature interaction module is introduced to enhance the information flow between different semantic levels, thus improving the recovery of small targets and edge details. In addition, for the problem of complex feature distribution in fog images, this paper introduces the ResBlock module with nested design in both encoder and decoder. Each ResBlock contains two 5×5 convolutional layers with a residual connecting path, which can more effectively model the spatial distribution characteristics of fog concentration by extending the receptive field to cover the typical fog interference region (about 30×30 pixels), compared with the traditional 3×3 convolutional structures [7]. It has been shown [15,45] that networks based on the encoder-decoder structure combined with multi-scale feature fusion can significantly improve the performance of image restoration and downstream detection tasks in low-contrast environments. Based on the above theoretical basis and the characterization of insulator images under complex meteorological conditions, this paper finally adopts the current FC-NET architecture and verifies its effectiveness and superiority in a complete system comparison experiment.

3.2.2. Feature fusion module

To achieve enhanced image de-fogging, a feature fusion method is devised to facilitate the comprehensive extraction of the features inherent to the input foggy image and the bootstrap image. Given that foggy images contain unprocessed information about the scene, including shape, colour and texture, extracting these features can assist

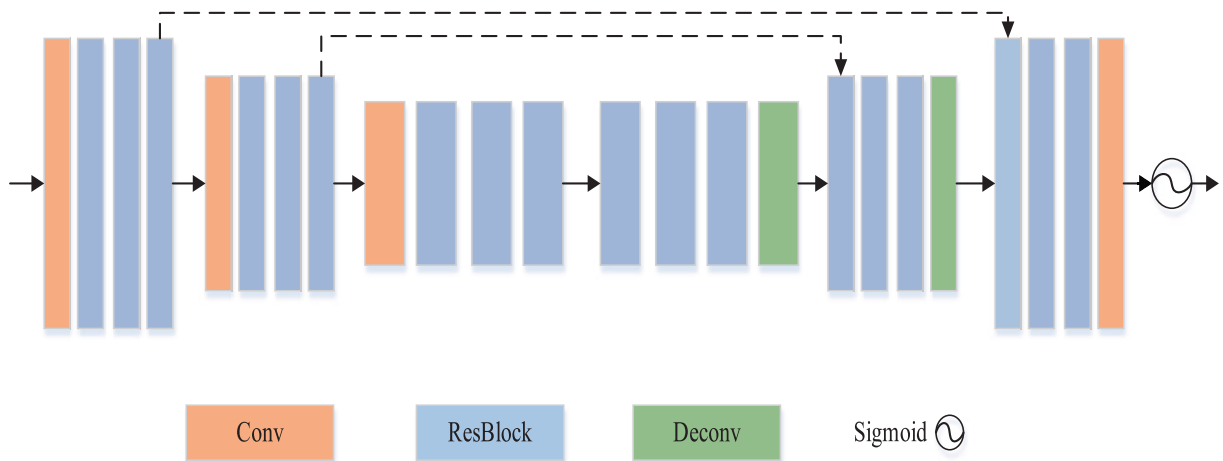


Fig. 4. Structure of the pre-fogging network.

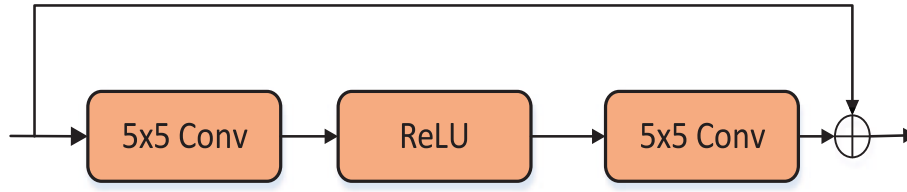


Fig. 5. ResBlock structure.

the model in comprehending the scene. The algorithm employs the feature extraction method $\mathcal{N}_i$, and following a convolution, introduces the Sandglass Block residual network for the extraction of features from foggy images. The detailed composition is shown in Fig. 6. It comprises two 3×3 deep convolutions and two 1×1 convolution blocks, which perform the operations of downscaling and upscaling, respectively. This effectively reduces the risk of information loss and gradient confusion, while enabling the learning of richer features.

The bootstrap image is subjected to feature extraction using the $\mathcal{N}_r$ method, which employs a 5×5 convolution and a ResBlock. Subsequent to this, the features F_I and F_R are obtained from the blurred and pre-misted images, respectively. The two features are then fused using the feature fusion method, with the initial features before fusion displayed Eqs. (5–7).

$$F_I^0 = F_I \quad (5)$$

$$F_R^0 = F_R \quad (6)$$

$$F_A^0 = S(F_I, F_R) \quad (7)$$

In this context, the notation $S(\cdot)$ represents the summation of two initial features, whereas F_A^0 denotes the features resulting from the initial aggregation of F_I and F_R . Subsequently, the aggregated features are subjected to further optimisation through a feature fusion technique, thereby incrementally enhancing the fused features, as shown in Fig. 7.

The feature fusion method comprises two modules: feature update and feature aggregation Eqs. (8–9). The feature update module is responsible for updating the fuzzy feature F_I^0 and the reference feature F_R^0 to obtain F_I^1 and F_R^1 .

$$F_I^1 = \mathcal{N}_g(F_I^0, F_A^0) \quad (8)$$

$$F_R^1 = \mathcal{N}_g(F_R^0, F_A^0) \quad (9)$$

The feature aggregation module performs a further aggregation of the updated features using a 3×3 convolution, before adding them with the initial aggregated feature, F_A^0 , to obtain the final fused feature, F_A as Eq. (10). The method fully exploits the characteristics of blurred and guided images to enhance the pre-fogging process.

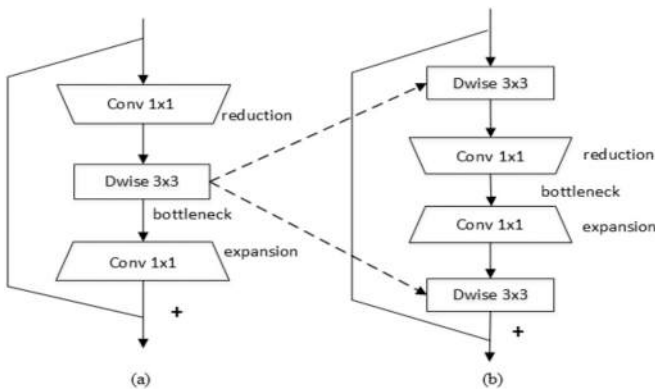


Fig. 6. Sandglass structure.

$$F_A = S(\mathcal{N}_i(F_I^1, F_R^1), F_A^0) \quad (10)$$

3.2.3. Image recovery module

In FC-NET, feature fusion uses a multi-stage approach by aggregating and progressively optimizing the features of the bootstrap and haze input images to achieve more efficient dehazing. First, feature maps are extracted from each input image, and the haze input image $I(x)$ is extracted by a convolutional neural network (CNN) to extract deep features to capture the details and global information of the image. Meanwhile, the features of the bootstrap map $R(x)$ provide additional information about the scene structure. In the feature fusion module, features from different inputs are combined using weighted aggregation and splicing operations, and the fused features are iteratively refined using a progressively optimized update-aggregation scheme to improve the consistency of global and local structures. The fused features are fed to the restoration module, where the correction coefficients $C(x)$ of the images are directly estimated according to Eq. (2) for image restoration. The multi-layer design of FC-NET ensures that the complementary features of the two input images are fully exploited to achieve the best dehazing effect, improving the clarity and naturalness of the final output image, while enhancing the model's adaptive capability in different scenes and complex environmental conditions.

The FC-NET method proposed in this study significantly improves insulator image quality under hazy weather through feature fusion, enabling the original problem of blurring of key features due to haze to be solved. For example, in transmission line fault detection, traditional detection methods are difficult to accurately identify faults because haze may hide insulator surface cracks and damage traces, whereas FC-NET significantly improves the reliability of fault detection by enhancing image clarity so that crack details are more easily captured.

3.3. Mixed scale based deraining method for transformer insulator images

The advent of CNN has led to the development of methods for removing rain from images. However, these methods are not well-suited to dealing with complex rain patterns due to the local receptive field and independence inherent to CNNs. The Visual Transformer was introduced into the field of image deraining, employing the self-attention mechanism to more effectively model non-local information. However, it has been observed that this approach may not fully address the challenges posed by high-frequency details. To consider both global and local features, this study combines Transformer and CNN, proposing a mixed-scale Transformer deraining method. This method captures features at different scales through a mixed-scale feed-forward network, enhances local feature recognition, and improves the deraining effect.

The overall architecture is based on a layered encoder-decoder design. The input is a rainy insulator data image $I_{rain} \in R^{H \times W \times 3}$, and the shallow feature $X_0 \in R^{H \times W \times C}$ is obtained by 3×3 convolution with a Leaky ReLU activation function. The shallow feature X_0 is then inputted into the encoder stage of the network, and each encoding layer contains a Transformer block and a downsampling operation. The Lewin Transformer [22] is employed as the underlying Transformer, and a mixed-scale feed-forward network is devised with the objective of capturing long-range dependencies, reducing the computational volume cost, and efficiently extracting local features at different scales. A 3×3

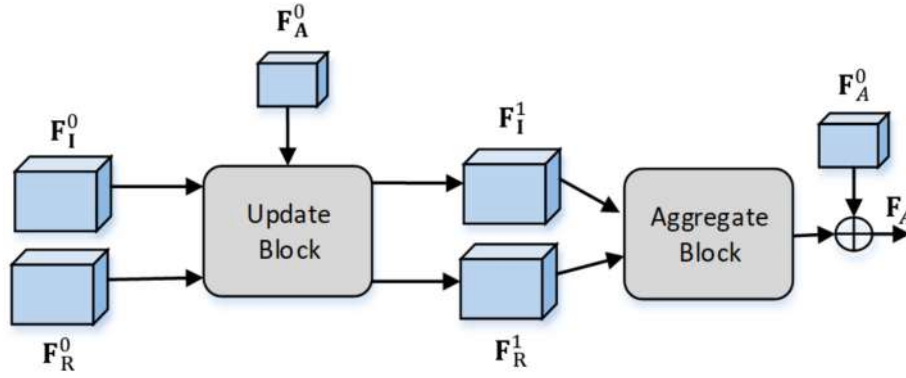


Fig. 7. Structure of feature fusion module.

convolution with a step size of 2 is employed for downsampling, resulting in a doubling of the number of channels and a halving of the height and width. The encoder outputs feature maps at varying levels through a process of stepwise downsampling. Feature reconstruction is then performed at the decoder stage, which is carried out by an encoder-like Transformer module and 3×3 upsampling with a step size of 2. The image dimensionality is subsequently recovered through a process of pixel shuffling. Jump connections are employed to integrate the encoder features with the decoder output, thereby facilitating image recovery. Finally, the feature refinement module enhances local feature recognition, obtains the residual map by 3×3 convolution and performs a jump connection with the input map I_{rain} to obtain the final deraining image I_{derain} . This is achieved through learning, as illustrated in Eq. (11). In these Eqs., $f(\cdot)$ denotes the overall deraining network, which has been designed to perform this process. The detailed structure of the deraining network is shown in Fig. 8.

$$I_{derain} = f(I_{rain}) + I_{rain} \quad (11)$$

3.3.1. Feature fusion techniques in FC-NET methods

The TFRAIN method introduces a novel hybrid scale transformer architecture designed to efficiently process rain streaks and combine local and global feature extraction mechanisms. Firstly, TFRAIN extracts deep features from rainy day images by means of a convolutional neural network (CNN), which are capable of efficiently capturing both detailed and global information of the image. In the encoder, the Windowed Multiple Self-Attention (W-MSA) mechanism splits the input image into non-overlapping image chunks to capture local contextual information while reducing computational complexity. The introduction of relative position coding further enhances the model's understanding of the

relationship between different pixel positions. Next, the Mixed Scale Feedforward Network (MSFN) combines 3×3 and 5×5 deep convolution to efficiently extract multi-scale features, and the extracted features are fused by splicing and residual joining to ensure that critical rain streak details are not lost during processing. The decoder structure then mirrors the encoder and reconstructs the de-raining image using an upsampling layer and a transformer module, while maintaining spatial detail through jump connections. The approach significantly improves the quality of the de-raining by combining the transformer (for modeling global dependencies) and the convolution (for preserving local details), enabling the model to efficiently process rain streaks of various scales, intensities, and textures, outperforming both traditional methods and CNN-based techniques.

The TFRAIN module, based on the Hybrid Scale Transformer, addresses the difficulty of detecting small target faults (e.g., micro-cracks on insulator surfaces, partial discharge traces) caused by raindrop occlusion by capturing multi-scale features and enhancing the ability to recover local details. This capability is particularly important for identifying small faults, especially in complex weather monitoring of transmission lines, and can effectively compensate for the distortion caused by raindrop interference in traditional detection methods.

3.3.2. Mixed-scale feed-forward network

The incorporation of a feed-forward network (FFN) serves to augment the model's capacity to effectively represent the input sequence through the utilisation of nonlinear transformation and feature extraction techniques. In the traditional approach to feed-forward networks, single-scale deep convolution is the predominant method for enhancing localisation. However, this approach fails to account for the correlation of multi-scale rain patterns. To enhance the image deraining

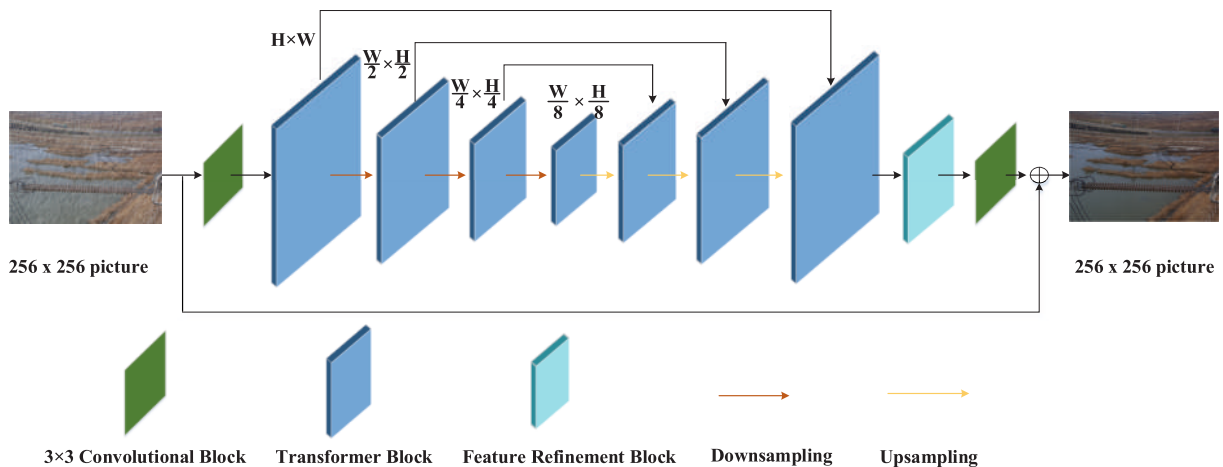


Fig. 8. Overall framework of the Going to the Rain Network.

effect, particularly in complex rainy scenes, this study proposes a mixed-scale feedforward network (MSFN) to augment the capacity for capturing multi-scale local information through the introduction of 3×3 and 5×5 multi-scale depth convolutions. The specific structure is shown in Fig. 9, wherein C represents the enhanced feature extraction.

In this mixed-scale feedforward network, assuming an input tensor $X_{l-1} \in \mathbb{R}^{H \times W \times C}$ is given, the channel dimensions are first extended using 1×1 convolution, and then the features are passed into the deep convolution to extract the multi-scale information using 3×3 and 5×5 convolution, respectively. Subsequently, the features of the two branches are channel spliced and summed with the features of the previous layer by 1×1 convolution. This process serves to preserve the original features and mitigate the loss of information, thereby forming the final output of the feedforward network. The complete process of feature extraction and fusion is illustrated in Eqs. (12–17).

$$\hat{X}_l = f_{1 \times 1}^c(\text{LN}(X_{l-1})) \quad (12)$$

$$X_l^{p1} = \sigma(f_{3 \times 3}^{dwc}(\hat{X}_l)) \quad (13)$$

$$X_l^{s1} = \sigma(f_{5 \times 5}^{dwc}(\hat{X}_l)) \quad (14)$$

$$X_l^{p2} = \sigma(f_{3 \times 3}^{dwc}[X_l^{p1}, X_l^{s1}]) \quad (15)$$

$$X_l^{s2} = \sigma(f_{5 \times 5}^{dwc}[X_l^{p1}, X_l^{p2}]) \quad (16)$$

$$X_l = f_{1 \times 1}^c[X_l^{p2}, X_l^{s2}] + X_{l-1} \quad (17)$$

In this context, $\hat{X}_l$ is used to denote the input of the current layer, whereas X_{l-1} is the output of the previous layer. The term $f_{1 \times 1}^c$ is used to denote a 1×1 convolution operation, while $f_{3 \times 3}^{dwc}$ and $f_{5 \times 5}^{dwc}$ are used to denote 3×3 and 5×5 deep convolution, respectively. The function $\sigma(\cdot)$ represents the ReLU activation function, and $[\cdot]$ are used to denote the channel connections.

The traditional Transformer is a computationally expensive approach due to the necessity of global computation of self-attention among all tokens, the difficulty of processing high-resolution images, and the single-scale deep convolution which ignores the multiscale correlation of rain patterns in rain maps. To address these issues, this paper proposes a novel hybrid multiscale Transformer block, which combines the hybrid scale feedforward network with the Lewin Transformer [22]. The detailed structure of this block is shown in Fig. 10.

Following the normalisation operation, the self-attention computation is conducted within each window utilising the Windowed Multiple Self-Attention Mechanism (W-MSA) module, thereby facilitating a more precise alignment with the local structural characteristics of the image. In the W-MSA module, the input 2D features are initially mapped into non-overlapping windows of size $M \times M$. Subsequently, the flattened and transposed features are obtained from each window, and the self-attention computation is conducted within each window. Let us as-

sume that the number of heads in multi-head attention is k and that the dimension of each head is $d_k = C/k$. In this case, the k -th head attention process in a non-overlapping window can be expressed as Eqs. (18–19).

$$Y_k^i = \text{Attention}(X^i W_k^Q, X^i W_k^K, X^i W_k^V), i = 1, \dots, N \quad (18)$$

$$X_k = \{Y_k^1, Y_k^2, \dots, Y_k^M\} \quad (19)$$

In this context, $N = HW/M^2$. The query, key and value projection matrices of the k th header, denoted by W_k^Q, W_k^K and W_k^V , respectively, are also considered. Additionally, X_k signifies the output of the k -th header across all windows. Ultimately, the outputs of all headers are merged through linear projection, resulting in the final outcome.

To facilitate a more nuanced comprehension of the interrelationships between disparate positions and to enhance the modelling capacity of the self-attention mechanism within a local window, a method of relative position encoding is employed in the attention calculation, as illustrated in Eq. (20).

$$\text{Attention}(Q, K, V) = \text{SoftMax}\left(\frac{QK^T}{\sqrt{d_k}} + B\right)V \quad (20)$$

In this context, the symbols Q, K , and V represent keys, values, and pairs, respectively. The symbol B denotes the relative positional bias. The utilisation of the relative positional bias matrix enables the model to more effectively consider the relative relationship between disparate positions, thereby facilitating a more precise capture of the pertinent feature information within the local window. It can be observed that the computational complexity of window attention varies from the original $O(H^2 W^2 C)$ to $O(M^2 HWC)$ for a given feature mapping input $X \in \mathbb{R}^{H \times W \times C}$.

A new hybrid multi-scale based Transformer block can be constructed based on the above window multi-head self-attention mechanism and mixed-scale feed-forward network. This new block helps the model to achieve feature fusion more efficiently and capture different scales and key information in the input features. In the case where the input feature of the $i-1$ th block is represented by X_{i-1} the expression of the i -th Transformer block is given by Eq. (21).

$$\begin{aligned} X'_i &= \text{W-MSA}(\text{LN}(X_{i-1})) + X_{i-1} \\ X_i &= \text{MSFN}(\text{LN}(X'_i)) + X'_i \end{aligned} \quad (21)$$

where $\text{LN}(\cdot)$ represents the normalisation operation, X'_i denotes the output of the input passing through the windowed multi-head self-attention mechanism, and X'_i is processed by the normalisation and the mixed-scale feed-forward network to obtain the final output X_i .

For the theoretical basis and task motivation considerations of the module selection. First, the Window Multi-Self-Attention (W-MSA) mechanism adopted by Lewin Transformer combined with the relative position coding strategy can model the global dependencies of images while keeping the computational complexity at a low level of $(O(M^2 N))$ while ensuring the global dependency modelling capability (where M is the window size and N is the sequence length), and can effectively ac-

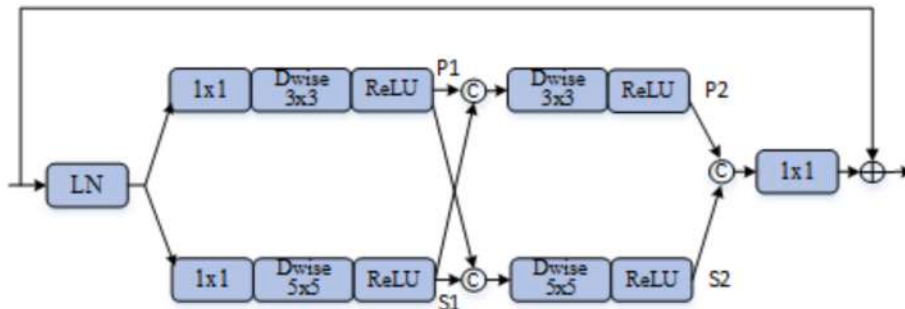


Fig. 9. Mixed-scale feedforward network.

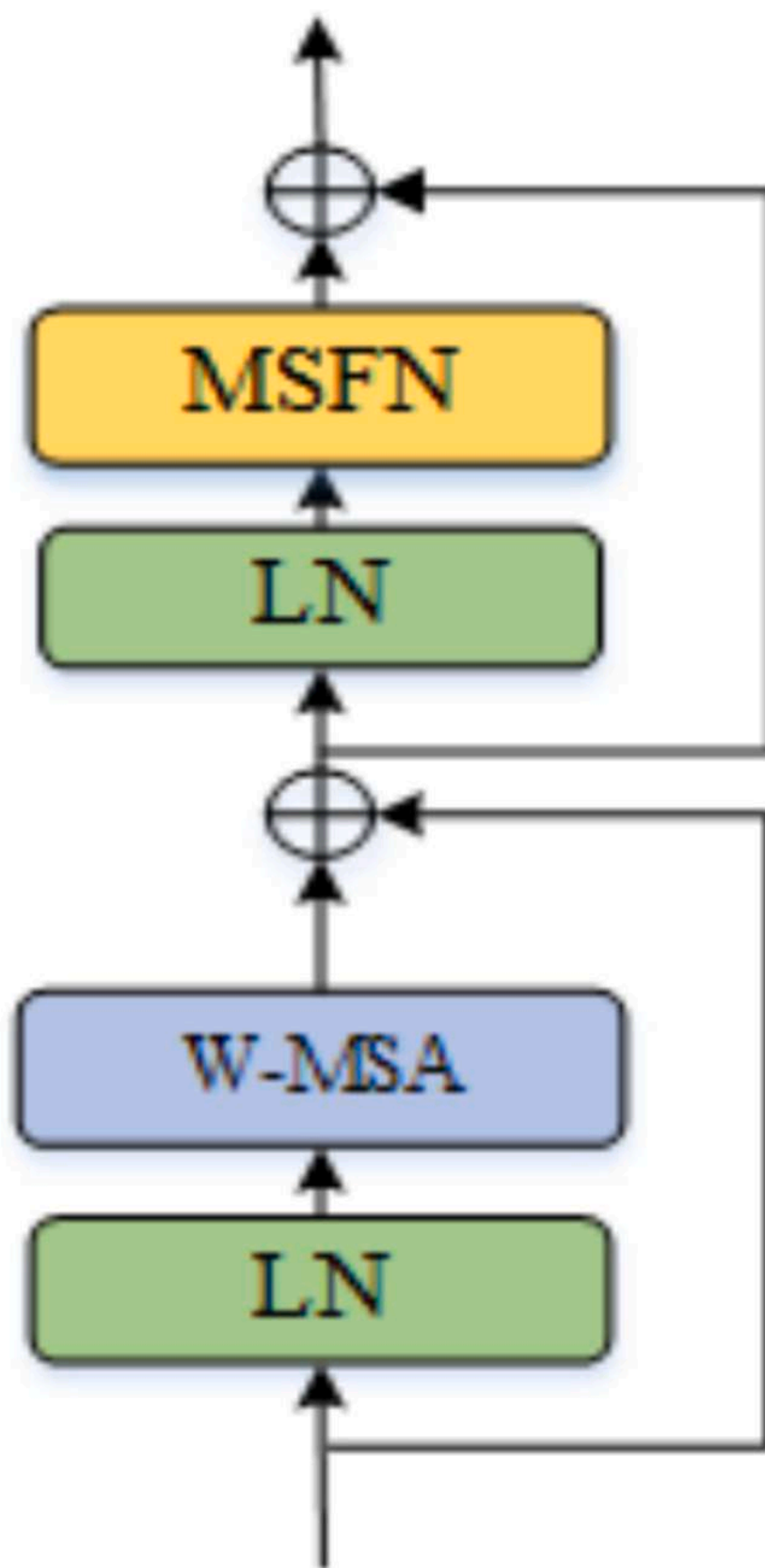


Fig. 10. Hybrid multi-scale Transformer block.

count for the global and local information balance [22]. Secondly, the feedforward network element introduces parallel 3×3 and 5×5 deep convolutional branches (MSFN). These draw on classic multi-scale structures, such as Inception [47] and Res2Net [48] with a view to enhancing the model's ability to model the details of complex degradation modes, including raindrops, water stains and mirror reflections, at different scales. The design has been developed to enhance the restoration of key features, such as texture and the edges of faulty regions, during image de-raining. This provides more discriminative input characterisation for the subsequent fault detection module in challenging environments, including those with low contrast and high blur. It can thus be concluded that the combination of Lewin Transformer and MSFN possesses not only theoretical structural complementarity and efficiency advantages, but also demonstrates the ability to adapt to the characteristics of the insulator detection task and complex meteorological scenarios.

3.4. A new lightweight high-precision target detection network

The principal algorithms for the detection of targets are classified into two principal categories. These algorithms can be divided into two categories: one-stage and two-stage. The YOLO series is an exemplar of the one-stage category, capable of completing detection in a single forward propagation and thus suitable for low-latency real-time applications. YOLOv7 is a widely used target detection algorithm that has been developed to improve accuracy and speed while enhancing scalability and versatility. However, as YOLOv7 is intended for deployment in a variety of practical applications, it must be capable of functioning effectively in complex detection environments, which present heightened demands in terms of accuracy and speed. Consequently, we have enhanced the YOLOv7 algorithm by incorporating the Efficient Multi-scale Attention (EMA) algorithm to reinforce the global contextual feature extraction and substituting the conventional convolution with the PConv convolution, which has enhanced the accuracy and lightweight performance of the model, thereby better meeting the requirements for insulator fault detection.

3.4.1. Add EMA attention

In YOLOv7-EP, the multi-scale attention mechanism (EMA) is introduced to enhance the model's ability to detect small targets in complex scenes. Although the traditional channel attention mechanism (CA) can enhance the weights of important features, it often fails to effectively capture details and contextual information when dealing

with small targets. And the EMA mechanism solves this problem by introducing multi-scale feature extraction and relative position coding to better capture the interrelationships between pixels, thus enhancing the detection of small targets. The EMA module first performs global average pooling on the input feature map to obtain global contextual information, and then extracts the local features through the convolutional layer. The introduction of relative position coding enhances the model's understanding of the relationships between different pixel positions, enabling the attention mechanism to aggregate features more effectively over channels and space. Cross-dimensional interactions play an important role in the prediction of channel or spatial attention. Based on this grouping structure, the EMA is interested in the CA Attention Mechanism (Coordinate Attention Mechanism), which represents an improved attentional mechanism for improving the performance of CNN. This mechanism incorporates spatial information into the channel attention, thereby enabling the model to capture more efficiently the input information that is important for the features. The sequential processing method has been enhanced to create a novel and efficient multi-scale attention algorithm that does not necessitate dimensionality reduction [49,50]. The model structure of CA and EMA is shown in Fig. 11.

The CA module performs a decomposition of the original input tensor into two parallel 1D feature-encoded vectors. Initially, two feature maps are generated and spliced along the vertical and horizontal directions through the implementation of two global average pooling of the input feature maps, which are then directed into the shared 1×1 convolution module. The feature map F_1 , which has undergone batch normalisation, is then fed into the feature map f , which has been obtained through the application of the Sigmoid activation function. Subsequently, f is split in both directions, resulting in the output vectors F_h and F_w , which collectively represent the features used for the perception of global location information. Subsequently, two attention weight vectors, g^h and g^w , are obtained through the application of a sigmoid function for the purpose of representing the relative importance of each location within the global information. Ultimately, the original input feature map is weighted and aggregated using the obtained attention weights, thereby producing the final output feature map Y .

However, the CA module fails to consider the interactions between spatial locations. Furthermore, the limited sensory field of the 1×1 convolution restricts the ability to model local cross-channel interactions, thereby underutilising contextual information and making it challenging to effectively capture complex spatial relationships. In EMA, based on the CA module, two parallel branches are employed to

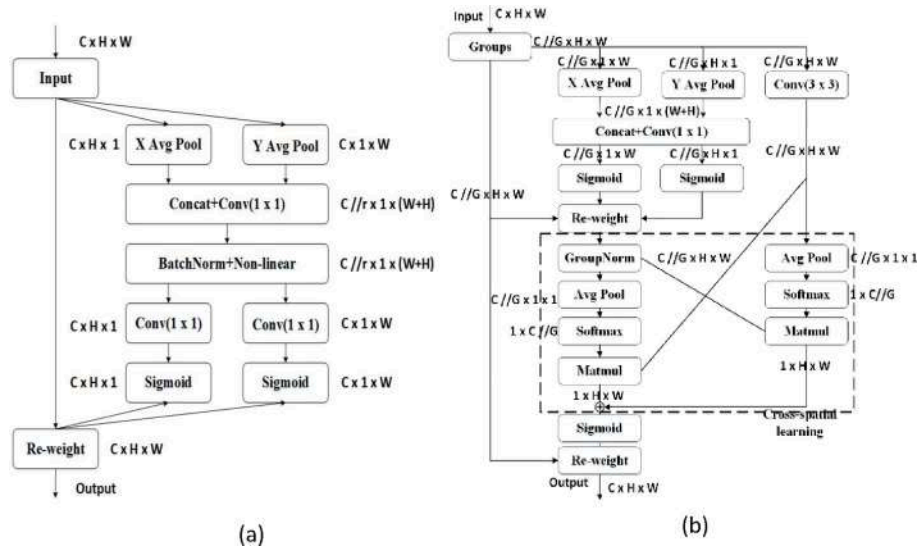


Fig. 11. Structure of CA attention (a) and EMA attention (b) models.

recalibrate the channel weights and further aggregate the output features through cross-dimensional interactions to capture pixel-level pairwise relationships, as illustrated in the structure of (b) in Fig. 11.

In the EMA model, the input feature mapping X is initially partitioned into G sub-features through the application of cross-channel dimension direction. The input feature mapping X is initially divided into G sub-features, and two additional 3×3 convolutions and one 1×1 convolution is incorporated to facilitate rapid response times. These are then modelled by two one-dimensional global mean pooling operations in the channel direction. In contrast to CA, EMA does not perform dimensionality reduction and employs two sigmoid functions to process feature encoding vectors, thereby enabling interaction between channels. The EMA module captures multi-scale features through 1×1 convolution and optimizes feature aggregation in the spatial dimensionality direction by encoding global spatial information using 2D global mean pooling. The design enhances the capacity to discern pairwise relationships at disparate channel and pixel levels, while also expanding the feature space. This, in turn, facilitates effective feature aggregation and more precise feature extraction. The expression is presented in Eq. (22).

$$z_c = \frac{1}{H \times W} \sum_j^H \sum_i^W x_c(i, j) \quad (22)$$

Subsequently, a SoftMax nonlinear function is applied to the output of the average pooling to accommodate the requisite linear transformation. Subsequently, the output of the parallel processing is multiplied with a matrix dot product operation, thereby obtaining the initial spatial attention map. In the 3×3 branch, the output is initially transformed into the format $C // G \times H \times W$, and subsequently, a second attention map is generated in a manner analogous to that employed for the 1×1 branch. Ultimately, the output feature maps within each group are aggregated to yield two sets of spatial attention weights. Following the application of the Sigmoid function and multiplication operations, the final output is of the same size as X and is efficiently embedded into the existing network structure.

An analysis of the YOLOv7 network reveals that the MP module employs pooling operations to reduce the feature map size, thereby alleviating the computational burden and extracting the primary features. However, this approach may result in the loss of crucial local features, potentially leading to inadequate attention to smaller targets. To enhance the role of the MP module in global contextual feature extraction, the MP module in the head of the YOLOv7 network is improved by introducing the EMA attention into the pooled convolution to form a new MP-EMA module. The resulting improved module structure is shown in Fig. 12.

3.4.2. Introduction of PConv convolution

Partial Convolution (PConv) is another important improvement in YOLOv7-EP, which aims to increase the efficiency of feature extraction. Conventional convolution often suffers from redundant computation when processing high-resolution images, leading to a waste of computational resources. PConv effectively reduces unnecessary computation by performing convolution calculations only on valid regions in the input feature map. Specifically, PConv operates only on non-zero inputs,

which improves the efficiency of feature extraction while maintaining high-quality feature output. This design allows PConv to reduce model parameters and computational complexity while still effectively capturing detailed features. PConv was first proposed by Chen et al [51], which possesses more efficient spatial feature extraction capability and less redundant computation and memory access requirements, and its basic working principle is shown in Fig. 13.

The distinctive aspect of PConv is its ability to extract spatial features through the implementation of regular convolution on a subset of the input channels, C_p , while the remaining channels remain unaltered. pConv employs a selective approach, utilising either the first or last C_p channels for computation, thereby ensuring the continuity of memory accesses. In the event that the input and output feature maps possess an identical number of channels, the number of floating-point operations for PConv is given by $h \times w \times k^2 \times c_p^2$, whereas the computation for regular convolution is represented by $h \times w \times k^2 \times c^2$, where c represents the number of channels present in the feature map. When the ratio $\frac{c_p}{c} = \frac{1}{4}$, the computation of PConv is only $\frac{1}{16}$ of that of ordinary convolution, while reducing the memory access. In this way, PConv can effectively extract spatial features while maintaining high efficiency and optimising the use of computational and memory resources.

To enhance the demand for real-time and lightweight processing in the insulator fault detection task, the ELAN module of the original network backbone is improved by replacing the 3×3 convolution operation in the module with the more lightweight PConv convolution to form the new ELAN-P module. The improved module is shown in Fig. 14.

By introducing the EMA attention mechanism and PConv convolution, the YOLOv7-EP module optimizes the ability to capture small targets and lightweight features, and is particularly suitable for the detection of small cracks, contamination and minor discharge faults common to trans Collusion line insulators. In complex contexts, the module can locate and classify these tiny fault targets more accurately, thus meeting the needs of practical applications. Based on the above improvements, this paper proposes a new type of target detection network, named YOLOv7-EP, which effectively improves the target detection capability under complex weather conditions by introducing EMA and PConv. These improvements enable YOLOv7-EP to maintain high accuracy while improving computational efficiency when dealing with different scenarios, providing new ideas and directions for future research. At the same time, the detection speed of the detection network is effectively eased to a certain extent. The structure of the enhanced YOLOv7 network is shown in Fig. 15, and the dashed boxes represent the enhanced modules of the network.

It is worth noting that the image processing and detection processes proposed in this study are well-combined. In the actual complex weather, the special scenario of fog and rain occurring at the same time. In this paper, a sequential processing mechanism is adopted, i.e., FC-NET is used to perform image fog removal, TFRAIN performs image rain removal, and finally inputs YOLOv7-EP to complete the fault detection, which forms a set of unified processing flow applicable to multiple weather disturbances.

4. Results

This paper introduces the evaluation metrics Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity (SSIM) for quantitative analysis. The PSNR is a metric used to assess the quality of an image. It is frequently employed to ascertain the degree of similarity between the original and processed images. A higher PSNR value indicates superior image quality, while a lower value indicates greater distortion. Its calculation expression is illustrated in Eq. (23).

$$PSNR = 10 \times \log_{10} \left(\frac{(2^n - 1)^2}{MSE} \right) \quad (23)$$

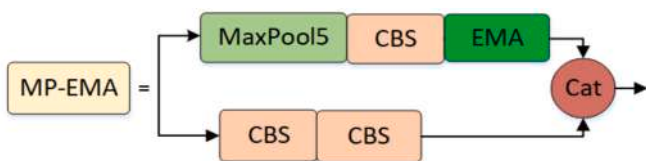


Fig. 12. Structure of MP-EMA module.

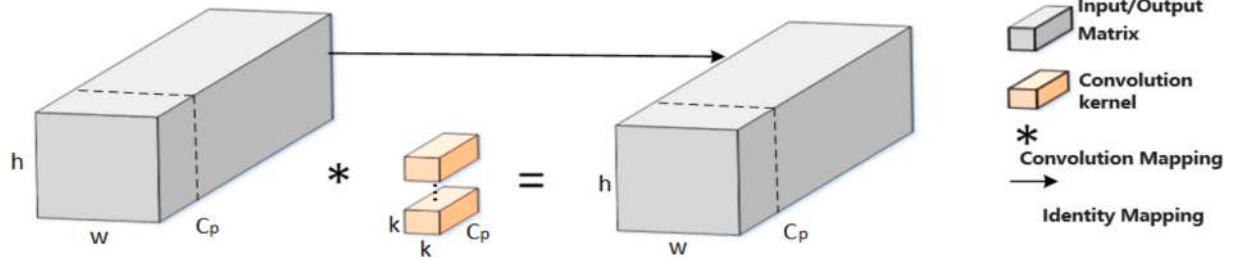


Fig. 13. PConv structure.

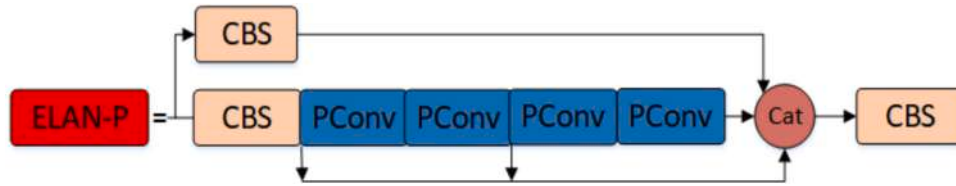


Fig. 14. Structure of ELAN-P module.

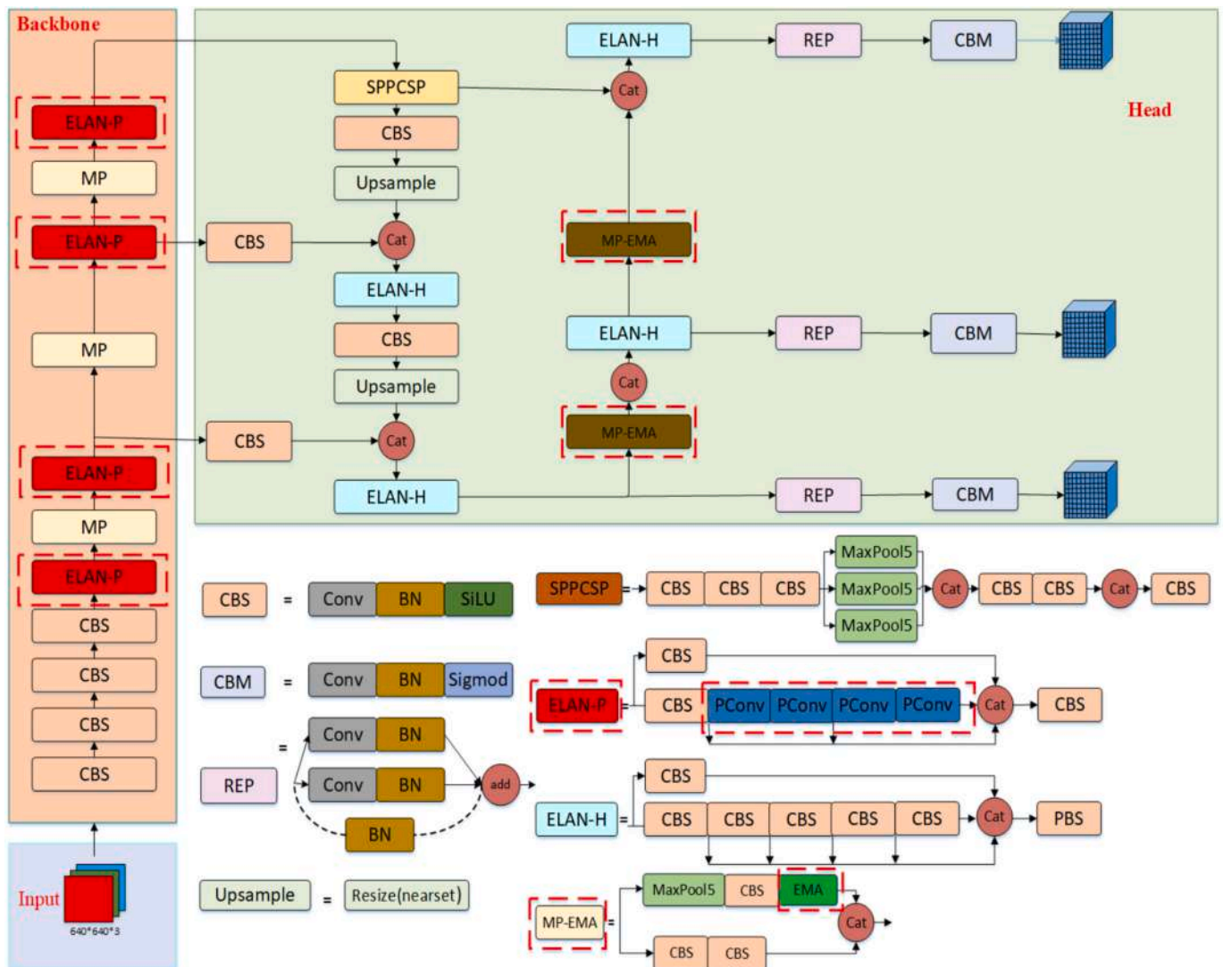


Fig. 15. Improved YOLOv7 network structure diagram.

where, MSE (mean square error) represents the average difference between the pixel values of two images.

SSIM can be employed as a means of quantifying the extent of SSIM between images. A combined analysis of the three aspects of information—brightness, contrast, and structure—provides a more comprehensive evaluation of image quality. The mathematical expression is provided in Eq. (24).

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (24)$$

where, μ_x and μ_y denote the mean of the two images, while σ_x^2 and σ_y^2 denote their variance. The notation σ_{xy} represents the covariance of the images, and the symbols C_1 and C_2 have been introduced to ensure that the denominator is not equal to zero. The value of SSIM ranges from -1 to 1 , with 1 indicating complete similarity between the two images, 0 indicating no SSIM, and -1 indicating complete contrast.

4.1. Feature fusion based insulator image dehazing experiment

To provide a comprehensive examination of the dehazing

performance of the feature fusion algorithm proposed in this paper. The existing dehazing algorithms, including the a priori dark channel dehazing algorithm DCP [32] and the network AOD algorithm based on deep CNN proposed by Li et al. [8], and the GCANet network [52] are evaluated through dehazing experiments based on insulator fogging data. The dehazing effects are assessed from both subjective and objective perspectives.

The network model is built under Ubuntu 18.04 system using Pytorch framework, and experiments are conducted on a GPU with 24G RAM model NVIDIA A40-24Q. The size of the input image is set to 256×256 , the number of batches is set to 8, and the epoch is set to 100 for FC-NET. The Adam's algorithm is taken as the optimiser, the momentum decay exponent $\beta_1 = 0.9$, $\beta_2 = 0.999$ and the initial learning rate is set to 1×10^{-4} .

4.1.1. Analysis of experimental results

The results of the subjective experiment conducted with the FC-NET and other dehazing algorithms are shown in Fig. 16. As can be observed from the figure, all the de-fogging algorithms utilised in the experiment have the effect of improving image clarity. The DCP algorithm is based on an a priori method, which has a poor de-fogging effect, with more haze residue and may lead to image distortion. In contrast, the AOD

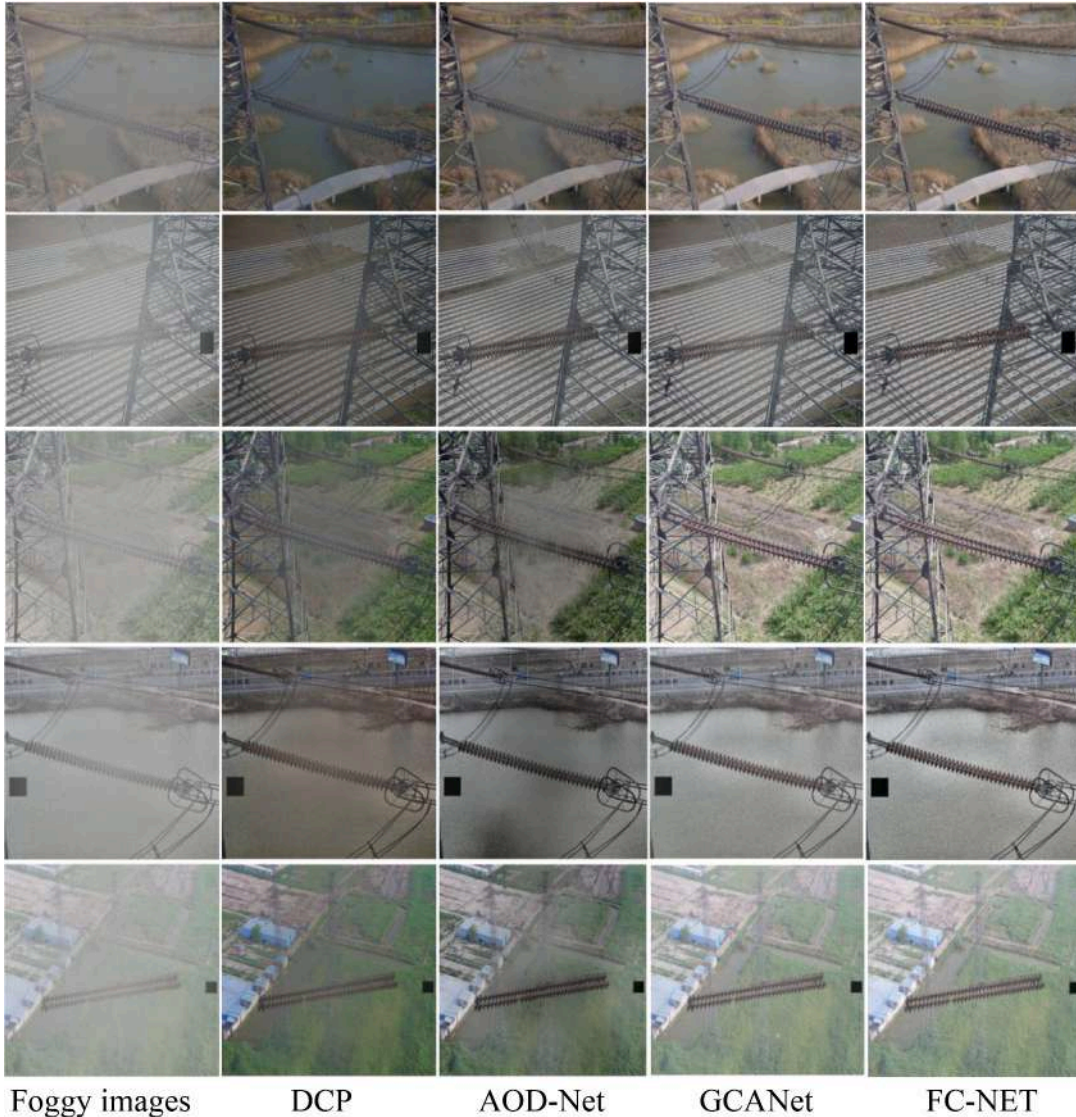


Fig. 16. Insulator fog map de-fogging effect comparison.

algorithm exhibits a superior dehazing effect; however, the resulting image is darker overall, with residual haze evident in the vicinity of the insulator. The GCANet algorithm demonstrates robust performance in dehazing insulator images, enhancing brightness while retaining some haze in complex backgrounds. Its detail recovery capabilities are generally satisfactory. The recently proposed dehazing method is capable of effectively removing the majority of haze, resulting in a colour closer to reality and a more effective detail recovery of the complex background.

As shown in Table 1, a comparative analysis of different dehazing algorithms is presented using two quantitative metrics, PSNR and SSIM, to assess their effectiveness. The results demonstrate that deep learning dehazing methods exhibit superior performance compared to traditional methods. The DCP and AOD-Net methods exhibit lower PSNR and SSIM values, indicating suboptimal dehazing efficacy and potential for image distortion. In contrast, the GCANet method demonstrates enhanced performance with a notable improvement. The FC-NET method, as proposed in this paper, has the highest PSNR and SSIM, with 2.41 dB and 0.022 improvements over GCANet, respectively. In conclusion, the algorithm in this paper demonstrates efficacy in both the dehazing effect and image detail recovery.

4.2. Experiments on deraining from transformer insulator images based on mixed scales

The experiments compare and analyse the TFRAIN method proposed in this paper with several common deraining algorithms, including the GMM [16] method based on a priori models, the deep learning method JORDER [53], based on multi-channel fusion, and the RESCAN [54] method based on deep convolutional and recurrent neural networks. Their deraining effects on the above datasets are evaluated from both subjective and objective dimensions.

The identical hardware environment utilized for the rain and fog method was employed for the training process. The Adam optimiser was employed, with the initial learning rate set to 1×10^{-4} , undergoing a gradual decrease to 1×10^{-6} during TFRAIN, and the number of batches set to 8, resulting in a total of 100 training sessions. In the experiment, images were randomly cropped into patches of 128×128 size. Additionally, a data enhancement strategy was employed whereby images were randomly rotated to increase the diversity of the training data.

4.2.1. Analysis of experimental results

This paper presents the results of deraining experiments conducted on synthetic insulator rain maps. The effects of various rain removal algorithms are demonstrated and analysed in comparison with each other from both subjective and objective perspectives. As shown in Fig. 17, the subjective visual impact of diverse rain removal algorithms on insulator synthetic rain maps is presented. The traditional GMM algorithm still leaves a considerable number of rain patterns after deraining. The overall effect of deep learning-based methods is superior to that of traditional algorithms. The JORDER and RESCAN algorithms are particularly effective in removing rain patterns, although a small number of rain patterns remain after the JORDER algorithm. Additionally, there are instances of chromatic aberration and artefacts on the background of the sky and the grass. The RESCAN algorithm demonstrates superior performance in deraining, yet it is inadequate for restoring details, resulting in blurring in certain scenes. In some

instances, the process of removal results in blurring. Following a subjective analysis, it was determined that the deraining algorithm presented in this paper exhibits the most optimal performance when applied to insulator data with rain. It is capable of effectively removing the rain pattern while simultaneously recovering the details.

As shown in Table 2, a comparative analysis of TFRAIN with the other three methods is presented using two performance indicators, PSNR and SSIM, to quantify the effectiveness of deraining. The specific data demonstrate that the traditional GMM method exhibits a low evaluation index and a poor deraining effect. In comparison, the evaluation indexes of the JORDER and RESCAN methods have been significantly improved in relation to GMM. Among these, the PSNR value of the RESCAN method is higher, indicating that it is more effective in recovering the image. The method proposed in this paper performs optimally under the same test environment, exhibiting a PSNR improvement of 2.32 dB and a SSIM value of 0.046 relative to the second-best RESCAN method. The quantitative comparative analysis results demonstrate that TFRAIN has a notable advantage in the task of removing rain from synthetic rain maps of insulators and has a higher value of application.

4.3. Insulator fault detection experiment under complex weather conditions

The experiments were conducted using a cloud server (The dehazing and deraining network and YOLO network were trained on two different configurations, A40-24Q and TRX2080ti, respectively, since cloud servers were used for training to prevent delays in the experimental process. Despite the different hardware configurations, both servers were tested to maintain the same training accuracy and model convergence when dealing with the same dataset and training parameters. In addition, deep learning models usually have good robustness and can obtain similar results under similar hardware conditions. Therefore, training with the A40-24Q and the TRX2080ti is equivalent, and the performance and validity of the models are reliable.), and the hardware and software environments are shown in Table 3.

The experimental dataset was selected from the expanded insulator dataset derived from the deraining experiment, comprising a total of 2544 sheets. Of these, 744 sheets were identified as defective insulator data, while the remaining 1800 sheets were classified as normal insulator data. From the aforementioned dataset, 2040 sheets were selected for training purposes, while 504 sheets were designated for testing. In addition, we performed multiple rounds of tuning and comparative analysis of key hyperparameters such as image input size, number of training rounds, batch size, learning rate strategy, etc., and finally determined the optimal performance of the configuration scheme. The input size is set to 640×640 , the number of training rounds is set to 200, the batch size is set to 8, the initial learning rate is set to 0.01, and the minimum learning rate is set to 0.00001.

4.3.1. Evaluation indicators

Insulator fault detection can be regarded as a categorical regression problem, and Average Precision (AP) is an important evaluation metric, which is calculated by the curves of Precision and Recall. Precision is used to evaluate the detection accuracy of the model, while Recall is used to evaluate the model's ability to detect the ability of all positive samples, which are calculated in Eqs. (25 and 26). In this experiment, TP denotes the number of faulty insulators successfully detected as faults, FP denotes the number of normal insulators mistakenly detected as faults, and FN denotes the number of insulators that failed to detect faults correctly. The AP can be obtained by integration calculation, and the average accuracy of all categories is further calculated and averaged to obtain the average accuracy value mAP of the whole dataset, which is shown in Eqs. (27 and 28).

Table 1
Objective comparison of insulator fog map de-fogging.

Arithmetic	PSNR	SSIM
DCP	16.75	0.814
AOD-Net	18.82	0.842
GCANet	23.40	0.903
FC-Net	25.81	0.925

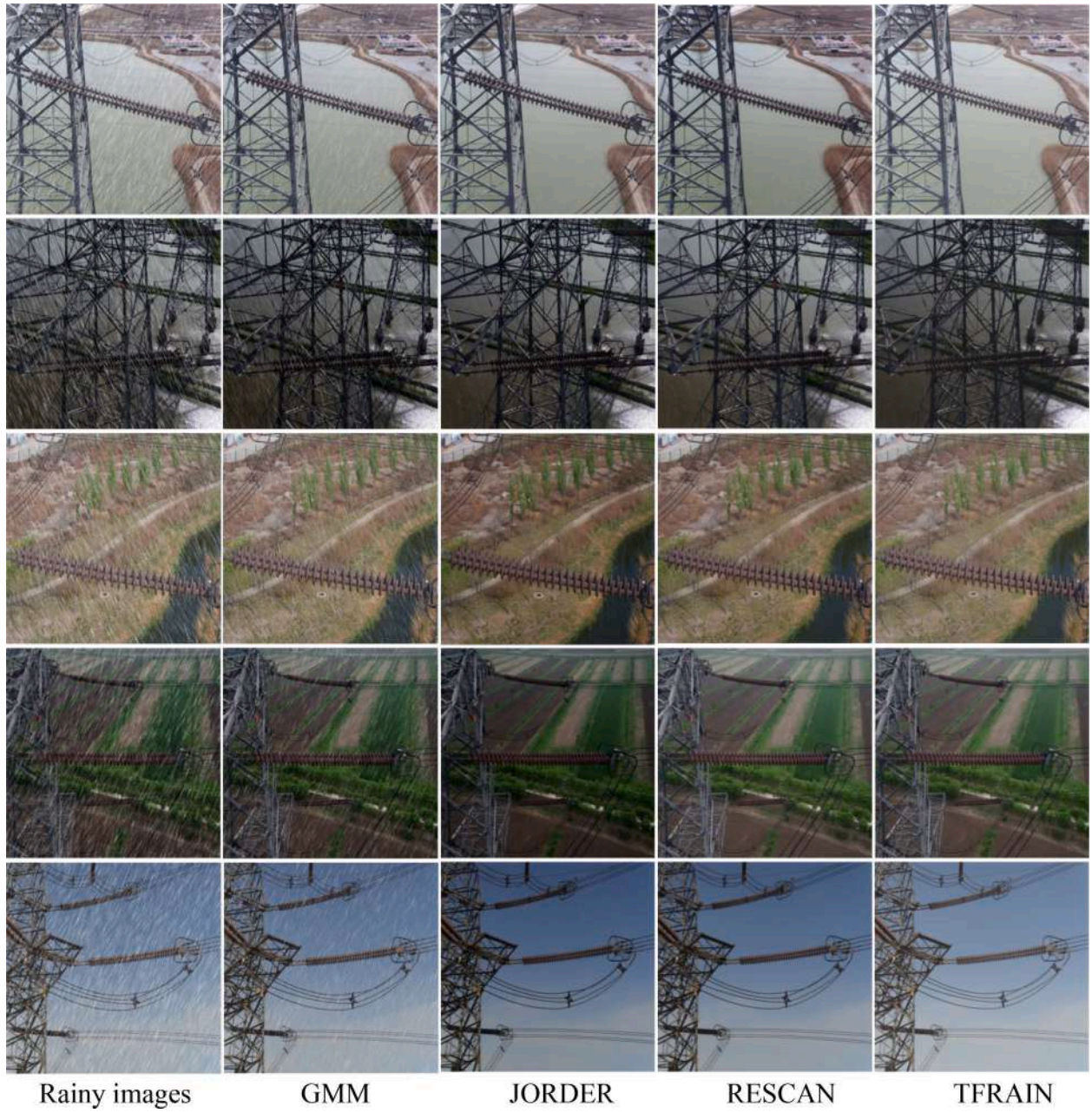


Fig. 17. Comparison of insulator synthetic rain map deraining effect.

Table 2
Insulator synthetic rain map deraining objective comparison.

Arithmetic	PSNR	SSIM
GMM	18.53	0.720
JORDER	24.28	0.823
RESCAN	27.06	0.875
TFRAIN	29.38	0.921

$$\text{Precision} = \frac{TP}{TP + FP} \quad (25)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (26)$$

$$AP = \int_0^1 PdR \quad (27)$$

Table 3
Experimental environment configuration.

Name	Configure
CPU	Intel(R) Xeon(R) CPU E5-2686
GPU	NVIDIA GeForce RTX 2080Ti
Operating System	Ubuntu 18.04
Memory	32G
Programming Language	Python3.8
CUDA Version	CUDA 11.6
Deep Learning Framework	Pytorch

$$MAP = \frac{AP_1 + AP_2 + \dots + AP_n}{n} \quad (28)$$

In the present experiments, Mean Average Precision (mAP) is utilised as the primary metric to evaluate the accuracy of detection. In more detail, mAP@0.5/% is utilised to evaluate performance at a constant

intersection over union (IoU) threshold of 0.5. Meanwhile, mAP@0.5:0.95/@ provides a more comprehensive assessment across multiple IoU thresholds (from 0.5 to 0.95 in steps of 0.05), aligning with industry standards such as the COCO evaluation protocol. A higher mAP is indicative of superior detection accuracy. Furthermore, the number of parameters (Params) reflects the spatial complexity of the network, while GFLOPs measure computational cost and frames per second (FPS) are reported as a key indicator of inference speed and real-time performance.

4.3.2. Analysis of experimental results

To explicitly analyse the independent contribution of each key enhancement module in the YOLOv7-EP network to the final detection performance, this study designed systematic ablation experiments to construct the YOLOv7-EMA vs. YOLOv7-PConv comparison models by eliminating the EMA attention mechanism and the PConv lightweight convolutional structure, respectively, and compared them with the Baseline (the original YOLOv7 model) and the full YOLOv7-EP. The experimental results are shown in Table 4, which quantifies the effects of each component on model recognition accuracy, computational complexity and number of parameters, respectively. The results show that the introduction of the EMA module alone improves the mAP @0.5/% by 0.9 %, effectively improving the detection of small targets while keeping the number of parameters unchanged; the replacement of the conventional convolution by PConv has a significant advantage in terms of lightweighting, with a reduction of 12.1 % in the number of parameters and 19.2 % in the amount of computation, despite a slight decrease in mAP @0.5/% of 0.6 %; and combining the two components to form YOLOv7-EP not only achieves 11.4 % of model accuracy, but also reduces the computational complexity of the model. The combination of the two into YOLOv7-EP not only achieves 11.6 % parameter compression and 18.3 % computational reduction, but also improves mAP @0.5/% by 0.7 % compared to the original mesh. A comparative analysis of the YOLOv7-EMA module with the baseline model reveals an enhancement of mAP@0.5:0.95 by 0.9 percentage points. This finding suggests that the employment of a weight update method based on moving averages has the potential to enhance the model's generalisation capability. While the YOLOv7-PConv module incorporating partial convolutional structures shows a slight decrease of 1.6 percentage points compared to the baseline model, it offers significant advantages in terms of parameter count and computational overhead. The complete YOLOv7-EP model, which combines EMA and PConv, achieves an mAP of 51.7 % while maintaining its lightweight characteristics. This indicates an enhancement of 1.5 percentage points in comparison with the baseline model and 3.1 percentage points in relation to the YOLOv7-PConv model. The ablation results confirm the complementary and synergistic value of the EMA and PConv modules in improving detection performance and model efficiency.

To In order to more comprehensively verify the performance advantages of the proposed target detection method, this paper conducts comparative experiments with the commonly used target detection models SSD, Faster R-CNN, DETR, YOLOv5, and YOLOX, and evaluates the performance of the different models in terms of precision, recall, and FPS, and so on. The results are given in Table 5. The results are shown in Table 5. Compared to SSD, Faster R-CNN and YOLOv5, the proposed

Table 4
Results of ablation experiments with EMA and PConv modules in YOLOv7-EP.

modelling	Parameters/ M	mAP @0.5/ %	mAP@0.5:0.95/ %	GFLOPs
Baseline	37.20	89.5	50.2	105.1
YOLOv7-EMA	37.21	90.4	51.1	105.3
YOLOv7-PConv	32.69	88.9	48.6	84.9
YOLOv7-EP	32.87	90.2	51.7	85.9

Table 5
Comparison of experimental results.

Model	Recall/%	Precision /%	FPS
SSD	76.2	85.2	53.9
Faster R-CNN	80.6	83.8	51.6
DETR	87.0	91.4	32.2
YOLOv5	83.7	84.1	59.7
YOLOX	87.3	86.2	58.9
YOLOv7-EP	86.6	92.6	61.3

algorithm improves the recall by 13.6 %, 7.4 % and 3.5 % respectively and the precision by 8.7 %, 10.5 % and 10.1 % respectively. Although slightly lower than YOLOX in terms of recall, precision is improved by 7.4 %. Compared to DETR, a target detection algorithm based on the Transformer architecture, the method in this paper is basically equal in terms of recall and precision. In terms of frame rate (FPS), compared to SSD, Faster R-CNN, DETR, YOLOv5 and YOLOX, the method in this paper improves by 13.7 %, 18.8 %, 90.4 %, 2.7 % and 4.1 % respectively. In summary, the algorithm proposed in this paper shows significant advantages in terms of completeness, accuracy and real-time processing capability in insulator fault detection, as shown in Table 5.

The FPS reported in Table 5 has been derived from single-threaded inference testing of the model on a desktop workstation. This was used to compare the relative performance of each algorithm under the same hardware conditions. As this study has not yet conducted experiments on embedded platform deployment, the results in the table are only provided as a reference for algorithm lightweighting and real-time performance improvements.

5. Discussion

5.1. Comparative insulator fault detection experiments under complex weather conditions

To validate the use of the de-fogging and deraining methods proposed in this paper, the insulator data collected under complex meteorological conditions can be processed to improve the accuracy of insulator fault detection, as well as the performance of using the improved YOLOv7 detection network in fault detection. Insulator fault detection experiments are carried out using the improved YOLOv7 network and the more excellent target detection networks Faster R-CNN [29] and SSD [55], respectively.

5.1.1. Foggy insulator fault detection

The integrated algorithm for insulator fault detection under foggy conditions is shown in Fig. 18. The structure is divided into two distinct sections: the training module and the detection module. In the training module, the dehazing model is trained using pairs of fogged images, and the target detection model is trained using the insulator dataset. The weights of the dehazing model and target detection model are then output, respectively. In the detection module, the dehazing model utilises the training weights to dehaze the insulator images, which are then input into the target detection network to achieve accurate fault detection. The integrated algorithm demonstrably enhances the reliability and accuracy of insulator fault detection in foggy conditions.

Based on the above integrated method of insulator fault detection under foggy conditions, the insulator detection effect before and after de-fogging is compared and analysed using the proposed improved YOLOv7 algorithm and the other two target detection algorithms, and the detailed results of the experiments are shown in Table 6.

As shown in Table 6, the mAP@0.5/% values of all three target detection algorithms in foggy conditions are lower than the detection results after processing using the dehazing network in this paper. Among them, the SSD algorithm has the lowest mAP@0.5/% value, while the YOLOv7-EP algorithm proposed in this paper shows the highest average

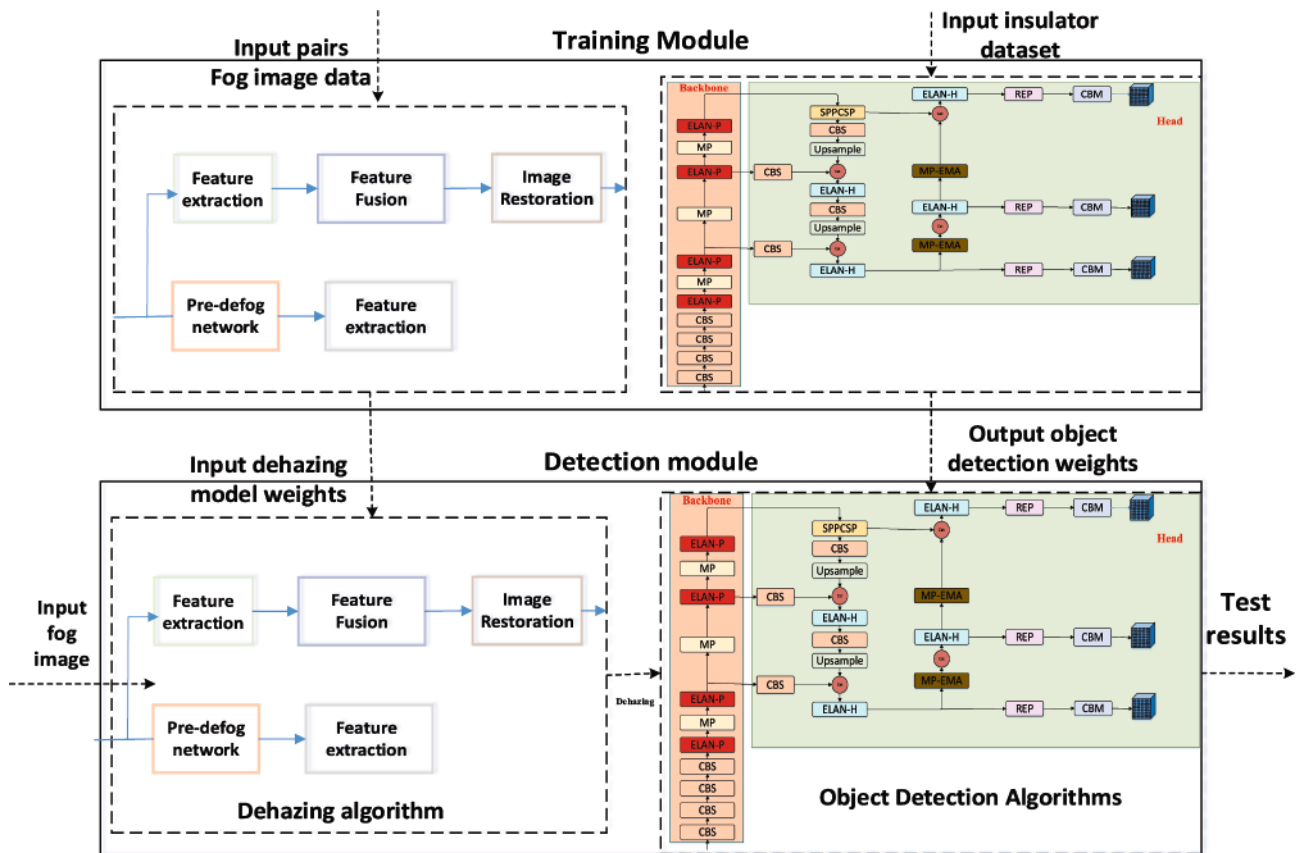


Fig. 18. Integrated algorithm for insulator fault detection under foggy conditions.

Table 6

Map@0.5/% before and after demisting treatment.

Contrast Methods	Faster R-CNN	SSD	YOLOv7-EP
Hazed image	84.5	81.5	88.7
Dehazed image	88.1	84.0	90.1

Table 7

Map@0.5:0.95/% before and after demisting treatment.

Contrast Methods	Faster R-CNN	SSD	YOLOv7-EP
Hazed image	45.2	42.8	51.0
Dehazed image	48.6	46.3	51.7

accuracy in both cases with fog and after dehazing. After the dehazing process, the average accuracies of the three models were improved by 3.6, 2.5 and 1.4 percentage points, respectively. The experiments show that the images after dehazing by the FC-NET method are not only significantly improved in terms of subjective clarity, but also show higher accuracy in detecting insulator cracks and discharge traces. For example, the accuracy of crack detection increased from 84.5 per cent to 90.1 per cent after the PSNR was increased by 2.41 dB, fully validating the module's enhanced ability to recognise specific fault types. In summary, the image de-fogging enhancement in this paper effectively improves the average accuracy of target detection and proves that the target detection algorithm in this paper shows better performance in insulator fault detection under foggy conditions.

As illustrated in Table 7, a comparative analysis has been conducted on the performance of three object detection algorithms in the context of hazy images and de-hazed images, with respect to the mAP@0.5:0.95 % metric. As demonstrated in Fig. 1, the mean accuracy of all three

detection models exhibited an enhancement following processing with the defogging algorithm proposed in this paper. The Faster R-CNN model demonstrated a 3.4 % enhancement, the SSD model exhibited a 3.5 % improvement, and the YOLOv7-EP model showed a 0.7 % enhancement. Of particular note are the enhancements observed in Faster R-CNN and SSD, suggesting a greater reliance on image clarity. Despite the marginal enhancement exhibited by YOLOv7-EP, its detection accuracy consistently surpasses that of the other two algorithms in both hazy and de-hazed contexts, thereby underscoring its notable robustness and stability.

Fig. 19 (a–d) illustrate the efficacy of the three target detection algorithms on insulators, both before and after processing with the de-fogging method proposed in this paper, (c) and (d) show an enlarged view of the faulty insulators located in the bottom right-hand corner of each image. The images are presented from left to right, respectively. The initial set of images depicts the detection effect of standard insulators, while the subsequent set illustrates the detection effect of faulty insulators. The results of the two groups demonstrate that, in conditions of low visibility, the accuracy of the three methods is relatively low, with the SSD algorithm exhibiting particular deficiencies. Following processing with the de-fogging algorithm, the detection accuracy of the three methods was enhanced, with the Faster R-CNN and SSD algorithms exhibiting particularly marked improvements. In both sets of detections, the various algorithms exhibited reduced accuracy for insulators situated at a distance, particularly in conditions of low visibility. Furthermore, the fog concentration in the second set of images is greater than in the first set, resulting in a lower overall detection rate than in the first set. The target detection network proposed in this paper yielded the most optimal results in both foggy and de-fogged data. The experiments demonstrated the efficacy of the de-fogging method proposed in this paper, and the target detection algorithm has greater potential for



(a) First set of foggy image detection results



(b) First set of dehazing image detection results



(c) Second set of foggy image detection results



(d) Second set of dehazing image detection results

Fig. 19. Detection effect before and after dehazing.

application in insulator fault detection in foggy scenarios.

5.1.2. Rainy day insulator fault detection

The structure of the integrated algorithm for insulator fault detection under rainy conditions is shown in Fig. 20 and is divided into two principal phases: the training module and the detection module. In the training module, the deraining model and the target detection algorithm are trained. The deraining model is trained using pairs of images captured on days with precipitation, while the target detection model is

trained using the insulator dataset. The training module generates the weights for the rain removal model and target detection model, which are then transferred to the detection module. In the detection module, the deraining process is performed on the rainy day images using the trained deraining model weights, thereby enhancing the clarity of the insulators. Subsequently, the processed image is fed into the target detection network, which accurately detects insulator faults. The integrated algorithm demonstrably enhances the accuracy and reliability of insulator fault detection in rainy conditions.

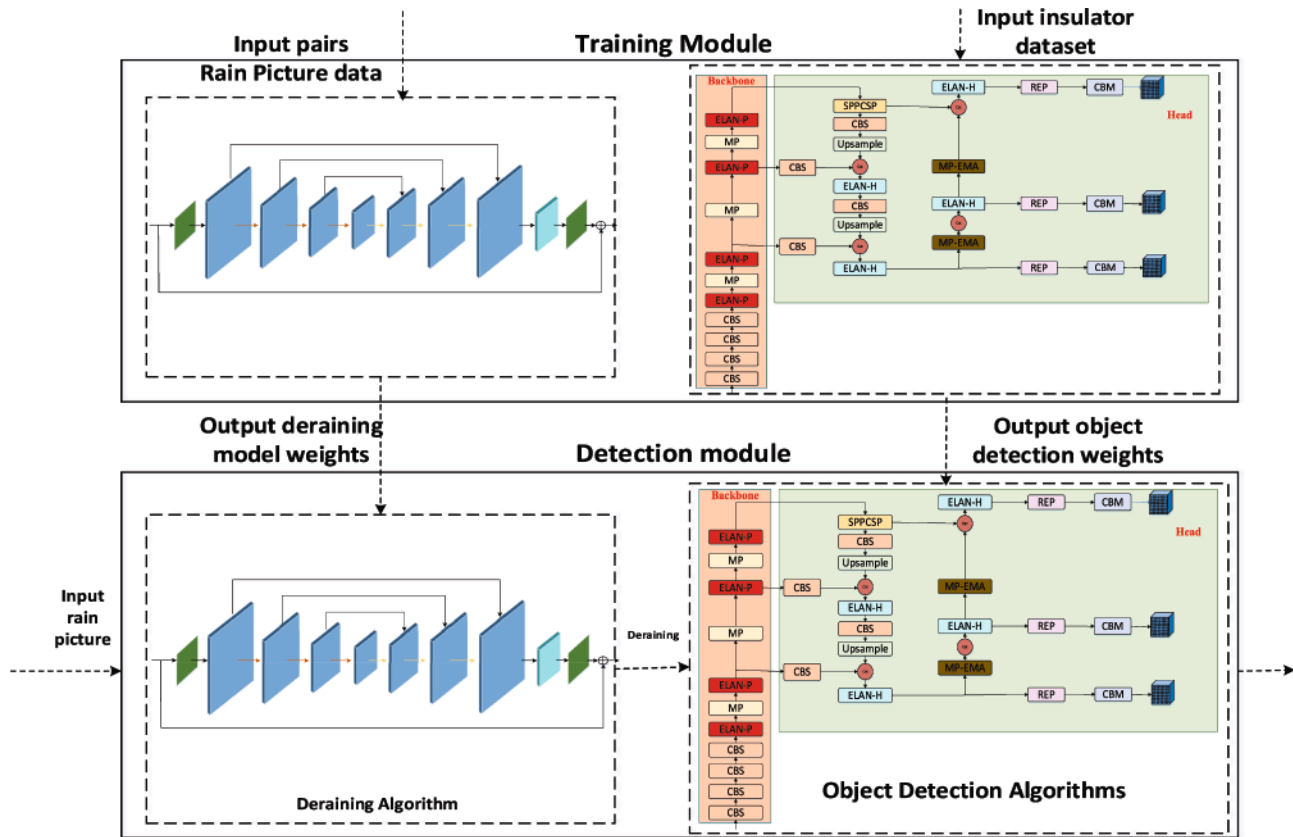


Fig. 20. Integrated algorithm for insulator fault detection under rainy conditions.

Based on the above integrated method for insulator fault detection under rainy conditions, the insulator detection effect before and after deraining is comparatively analysed using the improved YOLOv7 algorithm and the other two target detection algorithms, and the results are shown in Table 7.

The Table 8 illustrates that the $mAP@0.5/\%$ values of all three target detection algorithms have been enhanced for insulator detection by 2.2, 3.2, and 1.2 percentage points, respectively, through the utilisation of the deraining algorithm proposed in this paper. While the SSD algorithm with deraining has demonstrated a more pronounced enhancement, the resulting outcomes remain inferior to those of the other algorithms. In contrast, the newly proposed detection algorithm demonstrated the most optimal experimental outcomes in both detection with rain and subsequent deraining enhancement.

Table 9 shows the performance changes of three object detection algorithms before and after rain removal processing, as measured by the $mAP@0.5:0.95/\%$ metric. Using the rain removal algorithm proposed in this paper, the detection accuracy of Faster R-CNN, SSD, and YOLOv7-EP improved by 1.4, 3.5, and 0.5 percentage points, respectively. Among them, the improvement in the SSD algorithm was the most significant, indicating its sensitivity to rain interference; however, despite the substantial improvement, its overall performance still lagged behind the other two algorithms. Notably, the YOLOv7-EP algorithm achieved the highest detection accuracy on both rain-containing and rain-removed images, fully demonstrating its robustness and superior detection capabilities in complex weather conditions.

Table 8

$mAP@0.5/\%$ before and after deraining treatment.

Contrast Methods	Faster R-CNN	SSD	YOLOv7-EP
With rain image	86.1	82.2	89.0
Image after deraining	88.3	85.4	90.2

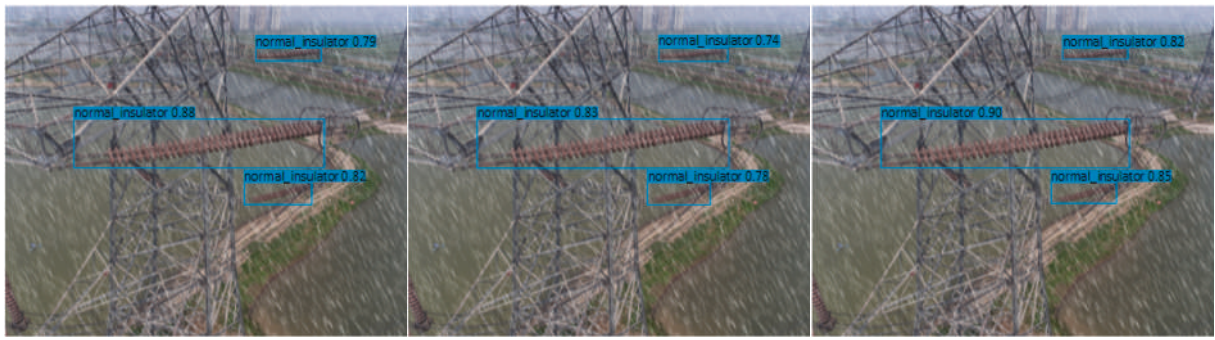
Table 9

$mAP@0.5:0.95/\%$ before and after deraining treatment.

Contrast Methods	Faster R-CNN	SSD	YOLOv7-EP
With rain image	47.5	44.1	49.3
Image after deraining	48.9	47.6	49.8

In rainy weather experiments, the TFRain module effectively eliminated rain drop interference, improving the detection network's $mAP@0.5/\%$ from 89.0 % to 90.2 % and $mAP@0.5:0.95/\%$ from 49.3 % to 49.8 % in small object recognition tasks. This result fully demonstrates the TFRain module's strong adaptability in small fault target detection tasks.

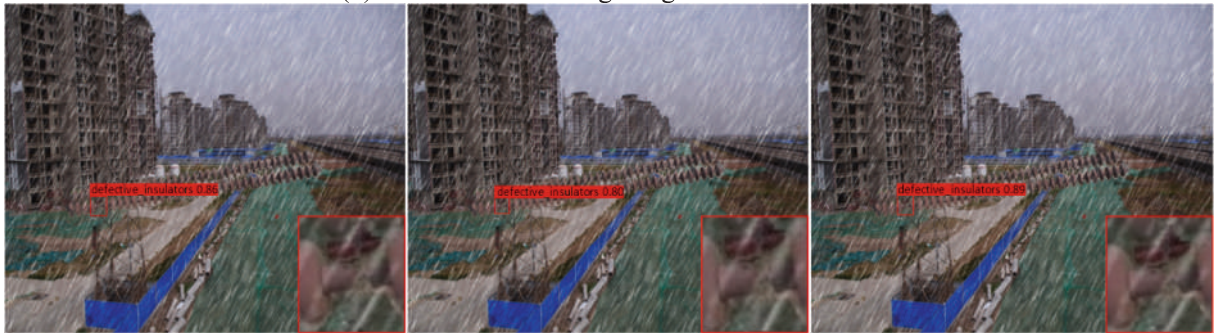
Fig. 21 (a–d) illustrates a comparison of the three target detection algorithms for insulators in scenarios with and without rain, (c) and (d) show an enlarged view of the faulty insulators located in the bottom right-hand corner of each image. The figure shows, from left to right, the effectiveness of the three algorithms in detecting insulators before and after the removal of rain, respectively. The first group illustrates the efficacy of the detection process for insulators in typical meteorological conditions, whereas the second group demonstrates the performance of the same process for insulators exhibiting defects in such conditions. As shown in Fig. 21, the three detection methods can identify insulator faults in both scenarios. However, following the implementation of the deraining technique proposed in this study, the detection accuracy is markedly enhanced, particularly for insulator targets situated at a distance from the target. The newly proposed target detection method demonstrates the most optimal results in both scenarios, thereby indicating its heightened potential for insulator fault detection under conditions of precipitation.



(a) First set of rainy image detection results



(b) First set of deraining image detection results



(c) Second set of rainy image detection results



(d) Second set of deraining image detection results

Fig. 21. Detection effect before and after deraining.

5.1.3. Comparative experiment summary

Considering the preceding two chapters' investigation into image de-fogging and deraining techniques, the effective identification of insulator faults in the presence of complex meteorological circumstances is now within reach. Firstly, a novel YOLOv7-EP target detection network is proposed, which is based on the YOLOv7 target detection network. This is achieved by introducing the EMA attention mechanism and replacing the traditional convolution with the PConv convolution.

Subsequently, two detection integration models are proposed for insulator fault detection under conditions of fog and precipitation. Finally, the efficacy of the de-fogging and deraining network treatments is validated through a comparative analysis of the detection outcomes on insulator datasets before and after the proposed modifications. The results demonstrate that the recently developed target detection algorithm exhibits an exceptional performance in detecting objects under both foggy and rainy conditions. Furthermore, the implementation of the de-

fogging and deraining algorithms in this study has been shown to markedly enhance the accuracy of insulator fault detection.

5.2. Theoretical contributions

The integrated technology of image processing and target detection under complex weather conditions is a key research direction in intelligent inspection systems. Based on the practical needs of insulator fault detection in power systems, this study proposes a series of innovative algorithms and models under complex weather conditions (foggy and rainy days), which significantly improve the accuracy and efficiency of insulator fault detection with theoretical contributions.

Firstly, an image dehazing method based on feature fusion is proposed to significantly improve the quality and detail fidelity of foggy day images. By constructing a feature fusion network (FC-NET), using a deep convolutional neural network to generate bootstrap images, and combining it with a progressive feature aggregation strategy, the performance bottleneck of the traditional fog removal method in complex scenes is overcome. Experimental results show that FC-NET improves PSNR and SSIM by 10.3 % and 2.4 %, respectively, on the insulator fog map dataset, and performs excellently in detail recovery and global feature enhancement.

Secondly, a mixed-scale transformer-based image de-raining method is designed to effectively solve the effect of rain interference on image quality. The proposed TFRAIN method combines a mixed-scale feed-forward network and a hierarchical encoding-decoding structure, which comprehensively improves rain pattern removal and detail recovery. Experiments show that the method significantly outperforms existing methods in terms of PSNR and SSIM metrics on synthetic rain map datasets, and achieves accurate extraction and fusion of multi-scale information.

Finally, a lightweight, high-precision object detection network, YOLOv7-EP, was developed to achieve efficient detection under complex weather conditions. By introducing an efficient multi-scale attention mechanism (EMA) and a PConv optimisation module, YOLOv7-EP significantly improved model performance in terms of small object detection and real-time performance. Experimental results demonstrate that the network improves detection accuracy (mAP) while reducing model parameters by 11.6 % and computational complexity by 18.3 %, providing robust support for insulator fault detection under complex weather conditions.

This study has important theoretical value and application potential in the field of image processing and target detection under complex weather conditions. The proposed multi-stage image processing and lightweight detection framework not only significantly improves the detection efficiency in complex scenarios, but also lays the foundation for the deployment of intelligent inspection systems in harsh environments.

5.3. Implications for practice

The method proposed in this paper has important application potential in many fields such as smart grid inspection, complex weather image processing and small target detection, and provides technical support and innovative ideas for related practices.

In recent years, power inspection under complex weather conditions has become a research focus, especially in the foggy and rainy environments where a breakthrough in image processing and fault detection problems are urgently needed. The feature fusion de-fogging method (FC-NET) and the mixed-scale Transformer de-raining method (TFRAIN) proposed in this paper significantly improve the image quality and detail fidelity under severe weather conditions. These techniques effectively solve the limitations of traditional methods with insufficient performance in dynamic environments, provide a reliable guarantee for the improvement of grid inspection efficiency and accuracy, and further promote the upgrading of UAV inspection technology to make the power

system operation more intelligent and efficient.

Meanwhile, the research in this paper has wide applicability in the field of UAV target detection. By adopting the lightweight YOLOv7-EP model, it can not only efficiently detect insulator faults, but also extend to application scenarios such as transmission line foreign object detection and critical equipment damage identification. These improvements reduce the dependence on manual detection, reduce the labour intensity and safety risk during inspection, and significantly improve the inspection efficiency and safety. In addition, the methods in this paper show a wide range of potential in the field of intelligent traffic monitoring. For example, combining the TFRAIN method to process data collected by traffic cameras in rainy days can significantly enhance image clarity and improve the accuracy of traffic congestion monitoring and licence plate recognition. This not only enhances the operational efficiency of the traffic management system, but also provides reliable data support for urban traffic planning and management.

In summary, the method proposed in this paper provides a systematic solution for image processing and small target detection under complex weather conditions, and improves the robustness and resource utilisation efficiency of the relevant systems. These research results are of great practical significance for power systems, intelligent transport and other intelligent fields that need to cope with complex environments, and further promote the application and development of related technologies in real-world scenarios.

5.4. Limitations and future research

As with all research, this work is not without limitations. There are still several open research questions that need to be further explored and solved. First, although the current pre-defogging network can extract effective features, its structure is relatively complex. Future research should explore lighter or more adaptive pre-processing methods to further improve inference efficiency while maintaining defogging quality. Second, this paper only verified fault detection on ceramic insulators. Future work should conduct experiments on various insulator types, such as composite and glass, to meet the needs of diverse inspection tasks. Third, although the mixed-scale Transformer (TFRAIN) achieves a good balance between performance and efficiency, there is still room for optimisation in its internal MSFN and window attention design. Future work could explore combining it with different Transformer architectures such as Restormer and Uformer, or introducing variants such as single-scale and group convolutions, to systematically evaluate the trade-off between computational overhead and visual quality. Meanwhile, due to the paucity of authentic adverse weather image samples, this study adheres meticulously to atmospheric scattering models and well-established schemes in the extant literature to synthesise fog and rain data. In the future, unsupervised domain adaptation and adversarial feature alignment techniques will be introduced to narrow the gap between the synthetic and real domains. Furthermore, physical modelling methods will be combined to construct cross-weather test sets for extreme weather conditions, such as snow and hail, to comprehensively validate the generalisation ability of the method. Finally, although YOLOv7-EP achieves high frame rates on desktop GPUs, deployment testing on edge devices such as Jetson TX2/Xavier NX after TensorRT FP16/INT8 quantisation and pruning is still required to measure FPS, single-frame latency, memory usage, and power consumption. This is to empirically demonstrate the model's real-time performance and feasibility in actual power grid inspection scenarios. The aforementioned work will establish a robust foundation for the development of a more efficient and general-purpose hybrid transformer-convolution detection framework.

From a holistic perspective, the reliability of a power transmission system is the result of the combined effect of several key components, including the electrical performance of the conductor and the mechanical stability of the support structure. Therefore, Future research can further extend the applicability of the module to cover more types of

faults, thus enabling more comprehensive intelligent detection of power transmission systems.

6. Conclusion

To address the issue of low insulator fault detection accuracy in the context of complex weather conditions, this paper puts forward a methodology for the removal of fog and rain from insulator images and combines it with target detection technology to develop an integrated model for the detection of faults in such conditions. Firstly, a feature fusion-based FC-NET de-fogging method is proposed to address the impact of foggy weather. A deep CNN is employed to construct a pre-fog removal model, and useful features are extracted from foggy images to generate guidance images. Subsequently, the extracted features from the foggy image and the bootstrap image are aggregated using a progressive feature fusion method. The image recovery module is constructed with the objective of performing image dehazing using the aforementioned fused features. The experimental results demonstrate that the method exhibits excellent performance in dehazing insulator images that have been affected by fog. Secondly, a mixed-scale Transformer (TFRAIN) deraining method based on mixed scales is proposed to address the issue of rain. Considering the significance of multi-scale information for the removal of rain patterns, a mixed-scale feed-forward network has been devised with the objective of enhancing the rain removal capability in complex scenes. The feature refinement module is incorporated into the feature reconstruction stage to facilitate the identification of local features within the model and enhance the effectiveness of deraining. The results of the comparative experiments conducted on the synthetic insulator rain map dataset demonstrate that the deraining algorithm can consider both global information and local details, which significantly enhances the deraining performance and provides robust support for insulator fault detection in complex meteorological conditions. In conclusion, the YOLOv7-EP target detection network, which incorporates EMA attention and PConv operation, has been proposed as a solution to the aforementioned issues. This network has been shown to enhance both the accuracy and efficiency of target detection, particularly in challenging weather conditions. The experimental results validate the effectiveness of this network in accurately identifying insulator faults in complex environments, comparisons were also made through FPS metrics to verify the real-time nature of the network. A comprehensive comparison and ablation experiment demonstrates that the method proposed in this paper can markedly enhance the accuracy of fault detection in both foggy and rainy conditions, and adequately evaluates the superiority of the proposed network through more diverse comparison experimental methods. This provides a robust technical foundation for the deployment of intelligent inspection techniques in power systems under complex weather conditions.

CRedit authorship contribution statement

Dong Jiang: Writing – review & editing, Writing – original draft, Supervision, Software, Data curation, Conceptualization. **Xiaoqiang Gong:** Writing – review & editing, Supervision, Investigation. **Chuang Yang:** Writing – original draft, Supervision, Resources, Methodology. **Bo Peng:** Supervision, Investigation. **Xiaojun Cheng:** Methodology, Investigation, Conceptualization. **Kanghua Yan:** Visualization, Supervision, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethics approval

Not applicable.

Consent to participate

Not applicable.

Consent for publication

All authors whose names appear on the submission approved the version to be published and agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Data availability

Data will be made available on request.

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